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Machine Design

4th Year Students

(Manual solutions)

Of machine design text book

by

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Solutions of Problems - Chapter 4

1- Given: $P = 65 \text{ kN} = 65 \times 10^3 \text{ N}$, $\tau = 35 \text{ N/mm}^2$, $d = ?$

$$\tau = P/A \Rightarrow 35 = 65 \times 10^3 / \left(\frac{\pi}{4} d^2 \right) \Rightarrow d = 48.6 \text{ mm, Say } d = 50 \text{ mm}$$

2- $\tau_{\max} = P_{\max}/A \Rightarrow 30 = \frac{580}{\frac{\pi}{4} d^2} \Rightarrow d = 4.9 \text{ mm} \sim 5 \text{ mm}$

3- $\tau_{\text{permissible}} = \frac{P}{A} \Rightarrow 52.5 = \frac{350 \times 10^3}{\frac{\pi}{4} d^2} \Rightarrow d = 92.13 \text{ Say } 95 \text{ mm}$

4- $\tau = \frac{P}{A} \Rightarrow 100 = \frac{5 \times 10^3}{\frac{\pi}{4} d^2} \Rightarrow d = 7.9 \text{ mm} \sim 8 \text{ mm}$

5- $\tau_{\text{bar}} = P/A_{\text{bar}} \Rightarrow P = \tau_{\text{bar}} \times A_{\text{bar}} = 150 (20 \times 20) = 60 \times 10^3 \text{ N}$

P for each bolt $= 60 \times 10^3 / 6 = 10,000 \text{ N}$

$$\tau_{\text{bolt}} = P_{\text{bolt}} / A_{\text{bolt}} \Rightarrow 75 = 10,000 / \left(\frac{\pi}{4} d^2 \right)$$

$$d = 13.029 \text{ mm say } 13 \text{ mm}$$

6- $P = P_r \times A = 0.7 \times \frac{\pi}{4} (300)^2 = 49.48 \text{ kN}$

$$\tau = P/A \Rightarrow 40 = 49.48 / \left(\frac{\pi}{4} d^2 \right) \Rightarrow d = 39.68 \sim 40 \text{ mm}$$

7- $\tau_c = P/A = 120 \times 10^3 / \left(\frac{\pi}{4} (50)^2 \right) = 61.115 \text{ N/mm}^2$

$$\tau = P/2A = 120 \times 10^3 / \left(2 \times \frac{\pi}{4} (40)^2 \right) = 47.75 \text{ N/mm}^2$$

8- $\tau_t = P/A \Rightarrow P = 1200 \times \frac{\pi}{4} d^2 = 300\pi d^2$

$$\tau = P/A = \frac{300\pi d^2}{\pi d t} = \frac{300\pi d}{\pi \times 20} \Rightarrow d = 20 \text{ mm}$$

9- $\tau_b = \frac{P}{L \cdot d} = \frac{35 \times 10^3}{1.2 d \times d} \Rightarrow 7 = \frac{35 \times 10^3}{1.2 d^2} \Rightarrow d = 64.55 \text{ mm}$

$$L = 1.2 d = 1.2 \times 64.55 = 77.46 \approx 80 \text{ mm}$$

$$10- A = \frac{\pi}{4} d^2 = \frac{\pi}{4} (20)^2 = 314 \text{ mm}^2$$

$$\sigma = P/A = 80000 / 314 = 254.77 \text{ N/mm}^2$$

$$\epsilon = \delta/L_0 = 0.048 / 40 = 0.0012$$

$$E = \sigma / \epsilon = 254.77 / 0.0012 = 212308.33 \text{ N/mm}^2 = 212.308 \text{ kN/mm}^2$$

$$\sigma_{\text{yield}} = \text{load at yield point} / A = 85 \times 10^3 / 314 = 270.56 \text{ N/mm}^2$$

$$\sigma_{\text{ultimate}} = \text{maximum load} / A = 150 \times 10^3 / 314 = 477.5 \text{ N/mm}^2$$

$$\text{Percentage elongation} = \frac{55.6 - 40}{40} \times 100 = 39\%$$

$$\text{Percentage reduction in area} = \frac{A_0 - A_r}{A_0} = \frac{20^2 - 12.4^2}{20^2} = 37.6\%$$

$$11- \text{case (1): } d_{s1} = 25 \text{ mm}, d_{b1} = 37.5 \text{ mm}, d_{b1} = 25 \text{ mm}$$

$$A_{s1} = \frac{\pi}{4} d^2 = \frac{\pi}{4} (25)^2 = 490.87 \text{ mm}^2, A_{b1} = \frac{\pi}{4} (d_b^2 - d_s^2) = 613.6 \text{ mm}^2$$

$$\sigma_{s1} = F/A_s = 10.186 \text{ MPa}, \sigma_{b1} = 8.15 \text{ MPa}$$

case (2), after turning:

$$d_{b2} = 25 + 8 = 33 \text{ mm}, A_{b2} = 364.6 \text{ mm}^2, A_{s2} = A_{s1} = 490.87 \text{ mm}^2$$

$$F_{s2} = F_{b2} \Rightarrow \sigma_{s2} \cdot A_{s2} = \sigma_{b2} \cdot A_{b2} \Rightarrow \sigma_{b2} = \frac{490.87}{364.6} \cdot \sigma_{s2}$$

$$\sigma_{b2} = 1.34 \sigma_{s2} \quad (1)$$

$$\delta l_{\text{net } s} = \delta l_{\text{net } b}$$

$$\delta l_{s2} - \delta l_{s1} = \delta l_{b2} - \delta l_{b1}$$

$$\left(\frac{P.L}{A.E} \right)_{s2} - \left(\frac{P.L}{A.E} \right)_{s1} = \left(\frac{P.L}{A.E} \right)_{b2} - \left(\frac{P.L}{A.E} \right)_{b1}$$

$$\frac{\sigma_{s2} - \sigma_{s1}}{E_s} = \frac{\sigma_{b2} - \sigma_{b1}}{E_b} \Rightarrow \frac{\sigma_{s2} - 10.186}{200 \times 10^3} = \frac{\sigma_{b2} - 8.15}{90 \times 10^3}$$

$$200 \sigma_{s2} - 1630 = 90 \sigma_{b2} - 916.74$$

$$200(1.34 \sigma_{s2}) - 1630 = 90 \sigma_{s2} \Rightarrow \sigma_{s2} = 3.98 \approx 4 \text{ MPa}$$

$$\therefore \sigma_{b2} = 3.98 \times 1.34 = 5.356 \text{ MPa}$$

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12- $\Delta t = t = 97 - 47 = 50^\circ\text{C}$

$$\delta_{al}(th) = \alpha \cdot t \cdot l = 24 \times 10^{-6} \times 50 \times 100 = 120000 \times 10^{-6} \text{ mm}$$

$$\delta_{st}(th) = \alpha \cdot t \cdot l = 12 \times 10^{-6} \times 50 \times 150 = 90000 \times 10^{-6} \text{ mm}$$

$$\delta_{net}(th) = (120000 - 90000) \times 10^{-6} = 30000 \times 10^{-6} \text{ mm}$$

$$\delta_{force(net)} = \delta_{al} + \delta_{st} = \left(\frac{P \cdot L}{A \cdot E} \right)_{al} + \left(\frac{(P - 200000) \cdot L}{A \cdot E} \right)_{st}$$

$$30000 \times 10^{-6} = \frac{P \times 100}{1000 \times 70 \times 10^3} + \frac{(P - 200000) \times 150}{1500 \times 210 \times 10^3} \Rightarrow P = 65750 \text{ N}$$

$$\sigma_{al} = P/A = 65750/1000 = 65.75 \text{ MPa}, \quad \sigma_{st} = 65750/1500 + \frac{200000}{1500} = 177.16 \text{ MPa}$$

13- $A_s = \frac{\pi}{4} d^2 = 314.16 \text{ mm}^2, \quad A_c = \frac{\pi}{4} (d_o^2 - d_i^2) = \frac{\pi}{4} (40^2 - 20^2) = 942.5 \text{ mm}^2$

$$\delta_c = \alpha_c \cdot t \cdot l = 18 \times 10^{-6} \times 50 \times l = 900 \times 10^{-6} l \text{ mm}$$

$$\delta_s = \alpha_s \cdot t \cdot l = 12 \times 10^{-6} \times 50 \times l = 600 \times 10^{-6} l \text{ mm}$$

$$\delta_{net}(th) = (900 - 600) \times 10^{-6} l = 300 \times 10^{-6} l \text{ mm}$$

$$\delta_{net}(force) = \left(\frac{P \cdot L}{A \cdot E} \right)_s + \left(\frac{P - 50000}{A \cdot E} \right)_c \cdot l$$

$$= \frac{P \cdot l}{314.16 \times 200 \times 10^3} + \frac{P \cdot l}{942.5 \times 100 \times 10^3} - \frac{50000 \times l}{942.5 \times 100 \times 10^3}$$

$$= 2.661 \times 10^{-8} \times P \cdot l - 53050.4 \times l \times 10^{-8}$$

$$\delta_{net}(force) = \delta_{net}(th)$$

$$2.661 \times 10^{-8} P \cdot l - 53050.4 \times l \times 10^{-8} = 300 \times 10^{-6} l$$

$$P = 31210.22 \text{ N}$$

$$\sigma_s = P/A_s = 31210.22/314.16 = 99.345 \text{ MPa} \quad \text{ans.}$$

$$\sigma_c = P/A_c = 31210.22/942.5 = 33.1 \text{ MPa}$$

$$\sigma_{camp} = F/A_c = 50000/942.5 = 53.5 \text{ MPa}$$

$$\sigma_{net} = 53.5 - 33.1 = 20 \text{ MPa} \quad \text{ans.}$$

$$14- \nu = E \epsilon, \epsilon = \nu / E, \epsilon = \frac{P}{A} / E = \frac{30 \times 1000}{20 \times 15 \times 200 \times 10^3}$$

$$\epsilon = 5 \times 10^{-4}$$

$$\epsilon_v = \epsilon(1 - 2\nu) = 5 \times 10^{-4} (1 - 2 \times 0.25) = 2.5 \times 10^{-4}$$

$$\epsilon_v = \delta v / V_0 \Rightarrow \delta v = V_0 \cdot \epsilon_v = 2000 \times 20 \times 15 \times 2.5 \times 10^{-4} = 150 \text{ mm}^3$$

$$V_0 = 2000 \times 15 \times 20 = 600 \times 10^3 \text{ mm}^3$$

$$V_{\text{final}} = V_0 + \delta v = 600 \times 10^3 + 150 = 600150 \text{ mm}^3 \text{ ans.}$$

$$15- \delta l = \frac{P \cdot l}{A \cdot E} \Rightarrow l = \frac{\delta l \cdot A \cdot E}{P} = \frac{3 \times \frac{\pi}{4} (12)^2 \times 200 \times 10^3}{8000} = 8482.3 \text{ mm}$$

$$\sigma_1 = \frac{W}{A} \left[1 + \sqrt{1 + \frac{2AEh}{W \cdot L}} \right]$$

$$= \frac{800}{\frac{\pi}{4} (12)^2} \left[1 + \sqrt{1 + \frac{2 \times \frac{\pi}{4} (12)^2 \times 200 \times 10^3 \times 80}{800 \times 8482.3}} \right] = 170.583 = 170.6 \text{ N/mm}^2 \text{ ans.}$$

Solutions of Problems - Chapter 5

$$1- \frac{T}{J} = \frac{\tau}{r} = \frac{G \cdot \theta}{L} \Rightarrow \frac{T}{J} = \frac{\tau}{r} \Rightarrow \tau = \frac{T}{J} \cdot r$$

$$\tau = \frac{1100 \times 1000}{\frac{\pi}{32} (d)^4} \times \frac{d}{2} \Rightarrow \tau = \frac{16 \times 1100 \times 100}{\pi (50)^3} = 44.8 \text{ MPa}$$

$$\frac{\tau}{r} = \frac{G \cdot \theta}{L} \Rightarrow G = \frac{\tau \cdot L}{r \cdot \theta} = \frac{44.8 \times 500}{25 \times 0.6 \times \frac{\pi}{180}} = 85.6 \text{ kN/mm}^2$$

$$2- T = \frac{P}{\omega} = \frac{100 \times 1000}{6\pi} = 5.31 \times 10^3 \text{ N.m} = 5.31 \times 10^6 \text{ N.mm}$$

$$\text{when } \omega = 2\pi N/60 = 6\pi \text{ rad/sec}$$

For strength:

$$\frac{T}{J} = \frac{\tau}{r} \Rightarrow \frac{5.31 \times 10^6}{\frac{\pi}{32} (d)^4} = \frac{60}{d/2} \Rightarrow d = 76.67 \text{ mm}$$

or

For stiffness:

$$\frac{T}{J} = \frac{G \cdot \theta}{L} \Rightarrow \frac{5.31 \times 10^6}{\frac{\pi}{32} (d)^4} = \frac{80 \times 10^3 \times 1 \times \frac{\pi}{180}}{3000} \Rightarrow d = 105 \text{ mm} \quad \text{ans}$$

for design we choose the greater, i.e. $d = 105 \text{ mm}$

$$3- T = \frac{P}{\omega} = \frac{90 \times 10^3}{6\pi} = 4.775 \times 10^3 \text{ N.m} = 4.775 \times 10^6 \text{ N.mm}$$

$$T_{\max} = T \times 1.4 = 4.775 \times 1.4 \times 10^6 = 6.684 \times 10^6 \text{ N.mm}$$

For strength

$$\frac{T}{J} = \frac{\tau}{r} \Rightarrow \frac{6.684 \times 10^6}{\frac{\pi}{32} (d)^4} = \frac{70}{d/2} \Rightarrow d = 78.6 \sim 80 \text{ mm}$$

$$\frac{T}{J} = \frac{G \cdot \theta}{L} \Rightarrow \frac{6.684 \times 10^6}{\frac{\pi}{32} (d)^4} = \frac{90 \times 10^3 \times \theta}{2000} \Rightarrow \theta = 0.037 \text{ rad}$$

$$\theta^\circ = 0.037 \times \frac{\pi}{180} = 2.116' \quad \text{ans}$$

$$4- \omega = \frac{2\pi N}{60} = 10\pi \text{ rad/sec}$$

$$T = \frac{P}{\omega} = \frac{11.2 \times 10^3}{10\pi} = 356.51 \times 10^3 \text{ N.m} = 356.51 \times 10^6 \text{ N.mm}$$

$$\frac{T}{J} = \frac{\tau}{r} \Rightarrow \frac{356.51 \times 10^6}{\frac{\pi}{32}(d_o^4 - d_i^4)} = \frac{80}{d_o/2}$$

$$d_i = \frac{3}{4} d_o$$

$$\frac{356.51 \times 10^6}{\frac{\pi}{32}[d_o^4 - (\frac{3}{4}d_o)^4]} = \frac{80}{d_o/2}$$

$$d_o = 321.4 \text{ mm}, d_i = 241 \text{ mm}$$

By correcting the results, we obtain:

$$d_o = 320 \text{ mm}, d_i = 240 \text{ mm} \quad \text{ans.}$$

$$5- V_{\text{solid}} = \frac{\pi}{4} d_o^2 \cdot l$$

$$V_{\text{hollow}} = \frac{\pi}{4} [d_o^2 - d_i^2] \cdot l = \frac{\pi}{4} [d_o^2 - (\frac{2}{3}d_o)^2] \cdot l$$

$$= \frac{\pi}{4} \cdot d_o^2 [1 - \frac{4}{9}] \cdot l = \frac{\pi}{4} d_o^2 \cdot \frac{5}{9} \cdot l$$

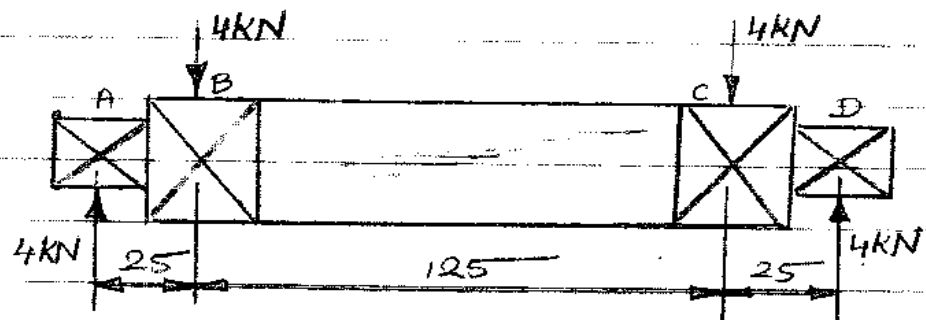
$$m = \rho V \Rightarrow W = \rho V g$$

$$W_{\text{solid}} = \rho \cdot V_{\text{solid}} \cdot g, W_{\text{hollow}} = \rho \cdot V_{\text{hollow}} \cdot g$$

$$\frac{W_{\text{hollow}}}{W_{\text{solid}}} = \frac{\rho \cdot V_{\text{hollow}} \cdot g}{\rho \cdot V_{\text{solid}} \cdot g} = \frac{V_{\text{hollow}}}{V_{\text{solid}}} = \frac{\frac{\pi}{4} \cdot \frac{5}{9} \cdot d_o^2 \cdot l}{\frac{\pi}{4} \cdot d_o^2 \cdot l} = \frac{5}{9}$$

$$= 0.56 \quad \text{ans.}$$

6-



Bending Moment at A = 0

$$= = = B = 4 \times 25 = 100 \text{ kN}\cdot\text{mm} \quad \curvearrowright$$

$$= = = C = 4 \times 150 - 4 \times 125 = 100 \quad \curvearrowright$$

$$= = = D = 0$$

∴ Maximum Bending Moment = 100 kN·mm (at B & C)

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$\frac{100 \times 10^3}{\frac{\pi}{64} d^4} = \frac{120}{d/2} \Rightarrow d = 20.4 \approx 21 \text{ mm}$$

$$7 \quad \omega = \frac{2\pi N}{60} = 10\pi \text{ rad/sec}$$

$$T = P/\omega = 20 \times 10^3 / 10\pi = 6.4 \times 10^3 \text{ N}\cdot\text{m} = 6.4 \times 10^5 \text{ N}\cdot\text{mm}$$

$$F = \frac{T}{R} = \frac{6.4 \times 10^5}{275/2} = 2.327 \times 10^3 \text{ N}$$

$$F_{\text{(for arm)}} = 2.327 \times 10^3 / 4 = 0.582 \times 10^3 \text{ N} = 582 \text{ N}$$

$$\text{Moment on each arm} = F_{\text{arm}} \times R = 582 \times 275 = 160050 \text{ N}\cdot\text{mm}$$

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$\frac{160050}{\frac{\pi}{4} \times a \times b^3} = \frac{15}{\frac{b}{2}}$$

$$b = 32.4 \text{ mm} \approx 32 \text{ mm} \quad \text{ans.}$$

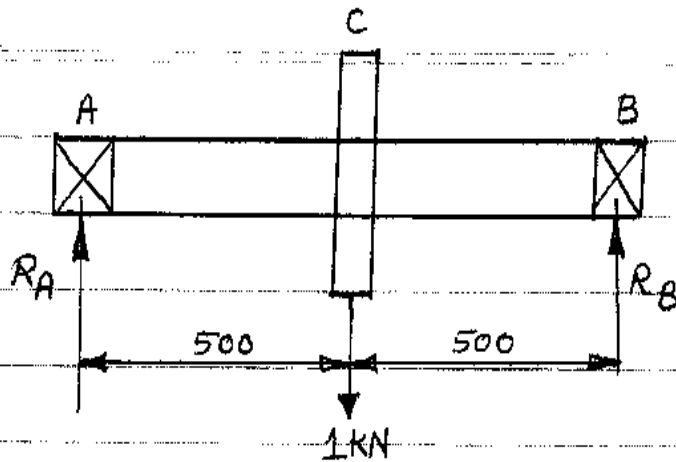
$$a = 2b = 2 \times 32 = 64 \text{ mm} \quad \text{ans.}$$

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8-

$$l = 1\text{ m} = 1000\text{ mm}$$

$$AC = CB = 500\text{ mm}$$



$$\sum M_B = 0 \Rightarrow R_A = 0.5 \times 10^3 \text{ N} = R_B$$

Bending Moment at point A = 0

$$C = R_A \times 500 = 500 \times 500 = 250 \times 10^3 \text{ N} \cdot \text{mm}$$

$$B = 0$$

Maximum bending Moment at the point C = $250 \times 10^3 \text{ N} \cdot \text{mm}$

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$\frac{250 \times 10^3}{\frac{\pi}{64} d^4} = \frac{40}{d/2} \Rightarrow d = 39.93 \approx 40 \text{ mm}$$

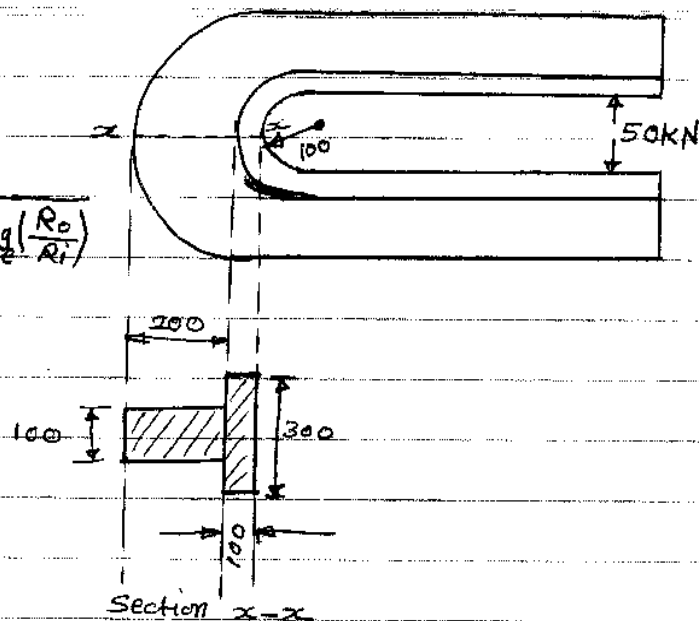
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g. By Using table 5.2

$$R_n = \frac{t_i(b_i - t) + t \cdot h}{(b_i - t) \cdot \log_e \left(\frac{R_i + t_i}{R_i} \right) + t \cdot \log_e \left(\frac{R_o}{R_i} \right)}$$

$$t_i = 100, b_i = 300, t = 100$$

$$h = 300, R_i = 100, R_o = 400$$



$$R_n = \frac{100(300 - 100) + 100 \times 300}{(300 - 100) \ln \left(\frac{100 + 100}{100} \right) + 100 \times \ln \left(\frac{400}{100} \right)}$$

$$= 180.337 \text{ mm}$$

$$R = R_i + \frac{\frac{1}{2} h^2 t + \frac{1}{2} t_i (b_i - t)}{h \cdot t + t_i (b_i - t)} = 100 + \frac{\frac{1}{2} (300)^2 100 + \frac{1}{2} (100)^2 (300 - 100)}{300 \times 100 + 100(300 - 100)} = 210 \text{ mm}$$

$$e = R - R_n = 210 - 180.337 = 29.663 \text{ mm}$$

$$y_i = R_n - R_i = 180.337 - 100 = 80.337 \text{ mm} \quad (\text{from } e \text{ to inner surface})$$

$$y_o = R_o - R_n = 400 - 180.337 = 219.663 \text{ mm}$$

$$x = x_o + R = 800 + 210 = 1010 \text{ mm} \quad (\text{the distance to C.A.})$$

$$A = 100 \times 300 + 200 \times 100 = 50000 \text{ mm}^2$$

$$\sigma_t = F/A = 50000/50000 = 1 \text{ MPa}$$

Bending Moment about C.A:

$$M = F \times x = 50000 \times 1010$$

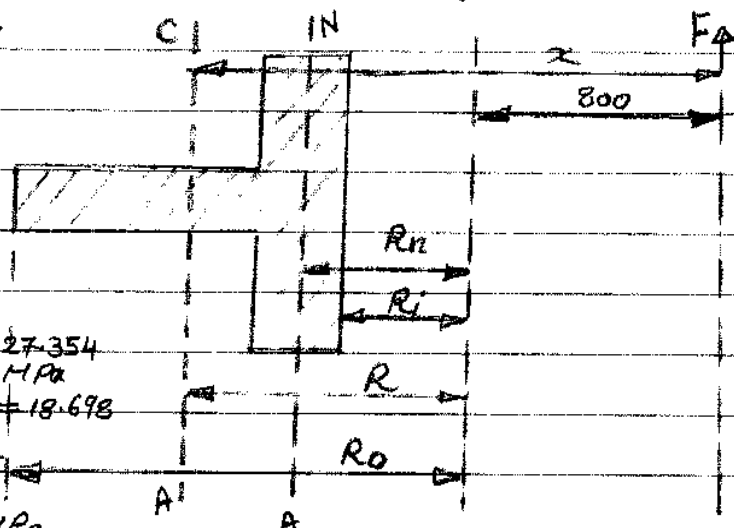
$$= 50.5 \times 10^6 \text{ N}\cdot\text{mm}$$

$$\sigma_{bi} = \frac{M \cdot y_i}{A \cdot e \cdot R_i} = \frac{50.5 \times 10^6 \times 80.337}{50000 \times 29.663 \times 100} = 27.354 \text{ MPa}$$

$$\sigma_{bo} = \frac{M \cdot y_o}{A \cdot e \cdot R_o} = \frac{50.5 \times 10^6 \times 219.663}{50000 \times 29.663 \times 400} = 18.698$$

$$\sigma_{Ri} = \sigma_t + \sigma_{bi} = 1 + 27.354 = 28.35 \text{ MPa}$$

$$\sigma_{Ro} = \sigma_t - \sigma_{bo} = 1 - 18.698 = 17.7 \text{ MPa}$$



10- $R_i = 65 \text{ mm}$, $R_o = 65 + 116 = 181 \text{ mm}$

$$A = \frac{1}{2} (90 + 25) * 116 = 6670 \text{ mm}^2$$

R_n = from table 5.2

$$= \left[\frac{b_i + b_o}{2} \right] h$$

$$= \left(\frac{b_i R_o - b_o R_i}{h} \right) * \ln \left(\frac{R_o}{R_i} \right) - (b_i - b_o)$$

$$= \left[\frac{90 + 25}{2} \right] * 116$$

$$= \left[\frac{90 * 181 - 25 * 65}{116} \right] * \ln \left(\frac{181}{65} \right) - (90 - 25)$$

$$= 103.48 \text{ mm}$$

$$R = R_i + \frac{h(b_i + 2b_o)}{3(b_i + b_o)}$$

$$= 65 + \frac{116(90 + 2 * 25)}{3(90 + 25)} = 112.072 \text{ mm}$$

$$e = R - R_n = 112.072 - 103.48 = 8.592 \text{ mm}$$

$$x = R = 112.072 \text{ mm}$$

$$M = P * x = 90 * 10^3 * 112.072 = 10086480 \text{ N.mm}$$

$$\sigma_e = P/A = \frac{90 * 10^3}{6670} = 13.493 \text{ N/mm}^2$$

$$y_i = R_n - R_i = 103.48 - 65 = 38.48 \text{ mm}$$

$$y_o = R_o - R_n = 181 - 103.48 = 77.52 \text{ mm}$$

$$\sigma_{b_i(p)} = \frac{M * y_i}{A * e * R_i} = \frac{10086480 * 38.48}{6670 * 8.592 * 65} = 104.193 \text{ MPa}$$

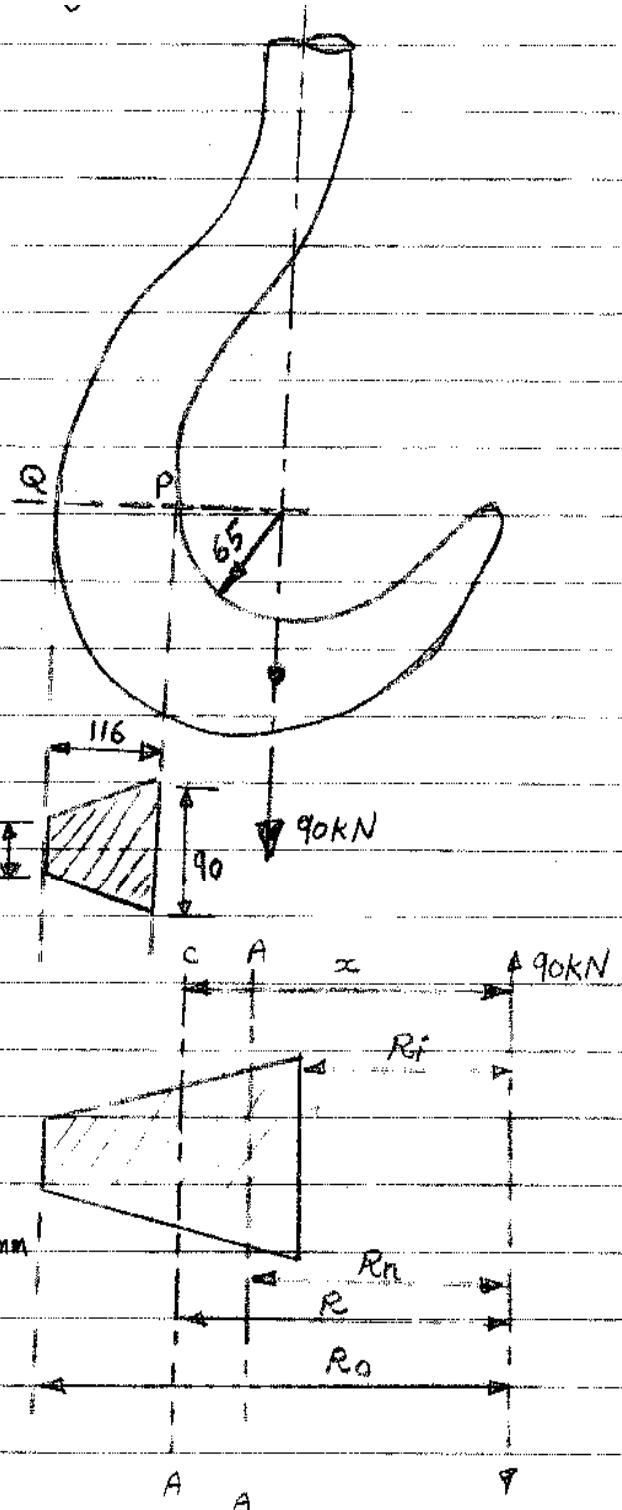
$$\sigma_{R_i} = \sigma_e + \sigma_{b_i} = 13.493 + 104.193 = 118 \text{ MPa}$$

$$\sigma_{b_o(t)} = \frac{M * y_o}{A * e * R_o}$$

$$= \frac{10086480 * 77.5}{6670 * 8.592 * 181} = 75.379 \text{ MPa}$$

$$\sigma_{R_o} = \sigma_e + \sigma_{b_o(t)} = 13.493 + 75.379$$

$$= 61.8 \approx 62 \text{ MPa}$$



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11- Given : $d = 16 \text{ mm}$, $P_t = 5000 \text{ N}$, $T = 50 \text{ N.m} = 50 \times 10^3 \text{ N.mm}$, $M = 75 \text{ N.m} = 75 \times 10^3 \text{ N.mm}$

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} (16)^2 = 200.96 \text{ mm}^2$$

(a) Max. stress theory:

$$\sigma_o = P/A = 5000/200.96 = 24.88 \text{ N/mm}^2$$

$$Z = \frac{\pi}{32} d^3 = \frac{\pi}{32} (16)^3 = 401.92$$

$$\sigma_b = M/Z = 75000/401.92 = 186.604 \text{ N/mm}^2$$

$$\tau = \frac{16T}{\pi d^3} = \frac{16 \times 50 \times 10^3}{3.14 (16)^3} = 62.20 \text{ N/mm}^2$$

$$\sigma_R = \sigma_o + \sigma_b = 186.604 + 24.88 = 211.484 \text{ N/mm}^2$$

$$\sigma_{\max} = \frac{\sigma_R}{2} + \frac{1}{2} \sqrt{(\sigma_R)^2 + 4\tau^2}$$

$$= \frac{211.484}{2} + \frac{1}{2} \sqrt{(211.484)^2 + 4(62.20)^2} = 228.421 \text{ N/mm}^2$$

$$\sigma_{\min} = -16.9373 \text{ N/mm}^2$$

$$F.S = \frac{\sigma_{yt}}{\sigma_{\max}} = 400/228.421 = 1.7511 \text{ ans.}$$

(b) Max. shear stress theory:

$$\tau_{\max} = \frac{1}{2} \sqrt{(\sigma_R)^2 + 4\tau^2} = \frac{1}{2} \sqrt{(211.484)^2 + 4(62.20)^2} = 122.63 \text{ MPa}$$

$$\tau_{\max} = \frac{\sigma_{yt}}{2 F.S} \Rightarrow F.S = \frac{400}{2 \times 122.63} = 1.63$$

We choose the greater value of $F.S = 1.7511 \text{ ans.}$

(2) By using distortion energy theory:

$$(\sigma_{e1})^2 + (\sigma_{e2})^2 - \sigma_{e1} \cdot \sigma_{e2} = \left(\frac{\sigma_{yt}}{F.S} \right)^2$$

$$(228.421)^2 + (-16.9373)^2 - (228.421) \cdot (-16.9373) = \frac{(\sigma_{yt})^2}{(1.686)^2}$$

$$\sigma_{yt} = 400.15 \approx 400 \text{ MPa.}$$

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$$F = 400 \text{ N}, \sigma_y = 320 \text{ MPa}$$

$$12- T = F \times 200 = 400 \times 200 = 8 \times 10^4 \text{ N}\cdot\text{mm}$$

$$\frac{T}{J} = \frac{\tau}{r} \Rightarrow \tau = \frac{T}{J} \cdot r$$

$$\tau = \frac{8 \times 10^4}{\frac{\pi}{32} d^4} \times \frac{d}{2}$$

$$= \frac{16 \times 8 \times 10^4}{\pi d^3}$$

$$B.M = F \times 400 = 400 \times 400$$

$$= 16 \times 10^4 \text{ N}\cdot\text{mm}$$

$$\frac{M}{I} = \frac{\sigma}{y} \Rightarrow \sigma = \frac{M}{I} \cdot y = \frac{16 \times 10^4 \times 32}{\pi d^3}$$

$$\sigma_{t1,2} = \frac{\sigma}{2} \mp \frac{1}{2} \sqrt{(\sigma)^2 + 4\tau^2}$$

$$= \frac{16 \times 10^4 \times 32}{\pi d^3} \mp \frac{1}{2} \sqrt{\left(\frac{16 \times 10^4 \times 32}{\pi d^3}\right)^2 + 4\left(\frac{16 \times 8 \times 10^4}{\pi d^3}\right)^2}$$

$$\sigma_{t1} = \frac{1726 \times 10^3}{d^3} \text{ N/mm}^2$$

$$\sigma_{t2} = \frac{-96.2 \times 10^3}{d^3} \text{ N/mm}^2$$

① For distortion energy:

$$(\sigma_{t1})^2 + (\sigma_{t2})^2 - 2\sigma_{t1} \cdot \sigma_{t2} = \left(\frac{\sigma_y}{F.S}\right)^2$$

$$\left(\frac{10^3}{d^3}\right)^2 \cdot [(1726)^2 + (-96.2)^2 - (1726 \times -96.2)] = \left(\frac{320}{4}\right)^2$$

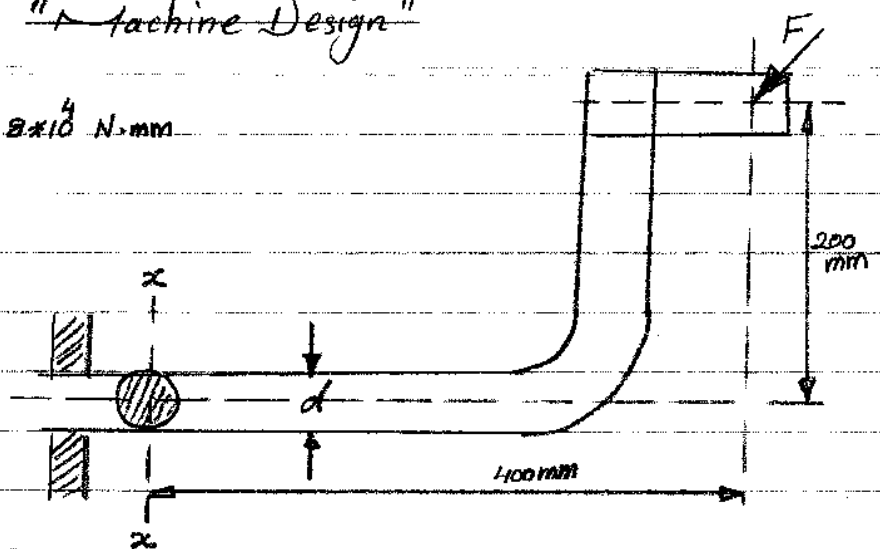
$$d = 28.36 \text{ mm} \quad \text{ans.}$$

② For Max. Shear Stress theory:

$$\tau_{\max} = \frac{1}{2} \sqrt{(\sigma_{t1})^2 + 4\tau^2} = \frac{1}{2} \sqrt{\left(\frac{16 \times 10^4 \times 32}{\pi d^3}\right)^2 + 4\left(\frac{16 \times 8 \times 10^4}{\pi d^3}\right)^2} = \left(\frac{\sigma_y}{F.S}\right)$$

$$= \frac{16 \times 10^4}{\pi d^3} \times \sqrt{320} = \frac{320}{8}$$

$$d = 28.34 \text{ mm} \quad \text{ans.}$$



13- $R_i = 100 - 50 = 50 \text{ mm}$, $d = 100 \text{ mm}$ Assume
 $R_o = 100 + 50 = 150 \text{ mm}$
 $R_n = \frac{(\sqrt{R_o} + \sqrt{R_i})^2}{4} = \frac{(\sqrt{150} + \sqrt{50})^2}{4} = 93.3 \text{ mm}$

$R = 100 \text{ mm}$

$e = R - R_n = 100 - 93.3 = 6.7 \text{ mm}$

$y_i = R_n - R_i = 93.3 - 50 = 43.3 \text{ mm}$

$y_o = R_o - R_n = 150 - 93.3 = 56.7 \text{ mm}$

$x = R + Z = 100 + Z$

$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} (100)^2 = 7854 \text{ mm}^2$

$\sigma_t = F/A = \frac{10 \times 10^3}{7854} = 1.273 \text{ MPa}$

Bending Moment at section A-A:

$M = F \times x = F(R + Z)$

$= 10 \times 10^3 (100 + Z) = 1000 \times 10^3 + Z \times 10 \times 10^3 \text{ N.mm}$

Max. Bending stress at inner fibre:

$\sigma_{bi} = \frac{M \cdot y_i}{A \cdot e \cdot R_i}$

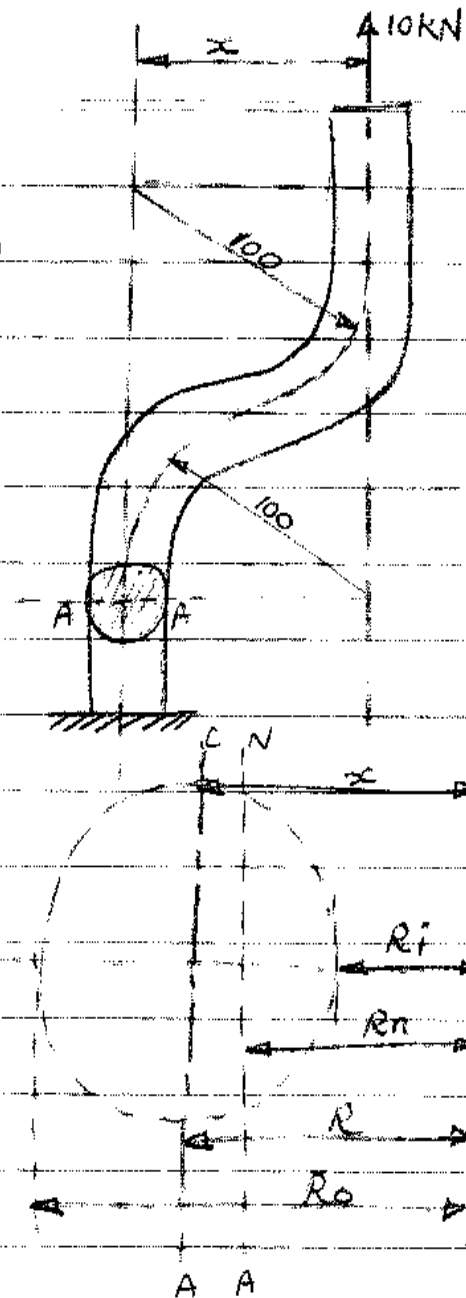
$= \frac{(1000 \times 10^3 + 10 \times 10^3 Z) \times 43.3}{7854 \times 6.7 \times 50} = 16.457 + 0.1645 Z$

Result stress at inside fibre:

$\sigma_t + \sigma_{bi} = 1.273 + 16.457 + 0.1645 Z = 17.73 + 0.1645 Z = 70$

$Z = 317.62 \text{ mm}$

$x = R + Z = 100 + 317.62 = 417.62 \approx 418 \text{ mm}$



$$14- A = \frac{\pi}{4} d^2 = \frac{\pi}{4} (50)^2 = 1963.5 \text{ mm}^2$$

$$R_n = \frac{(\sqrt{R_o} + \sqrt{R_i})^2}{4}$$

$$= \frac{(\sqrt{125} + \sqrt{75})^2}{4} = 98.4 \text{ mm}$$

$$R = R_i + \frac{d}{2} = 75 + \frac{50}{2} = 100 \text{ mm}$$

$$e = R - R_n = 100 - 98.4 = 1.6 \text{ mm}$$

$$y_i = R_n - R_i = 98.4 - 75 = 23.4 \text{ mm}$$

$$y_o = R_o - R_n = 125 - 98.4 = 26.6 \text{ mm}$$

$$\sigma_e = \frac{W}{A} = \frac{2 \times 10^3}{1963.5} + \frac{12 \times 10^3}{1963.5} = 7.13 \text{ MPa}$$

$$x = R + 75 = 100 + 75 = 175 \text{ mm}$$

$$M = W \times x = 2 \times 10^3 \times 175 = 350 \times 10^3 \text{ N.mm}$$

$$\sigma_{bi} = \frac{M \cdot y_i}{A \cdot e \cdot R_i}$$

$$= \frac{350 \times 10^3 \times 23.4}{1963.5 \times 1.6 \times 75} = 34.76 \text{ MPa}$$

$$\sigma_{bo} = \frac{M \cdot y_o}{A \cdot e \cdot R_o}$$

$$= \frac{350 \times 10^3 \times 26.6}{1963.5 \times 1.6 \times 125} = 23.71 \text{ MPa}$$

$$\sigma_e = \frac{W}{A} = \frac{2 \times 10^3}{1963.5} = 1 \text{ MPa}$$

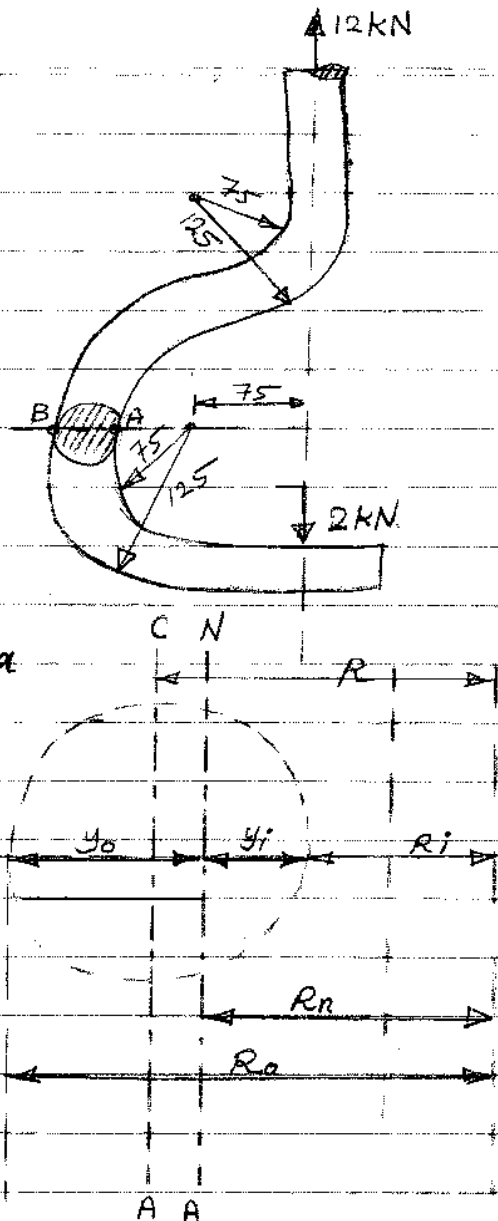
∴ Resultant stress at inner surface: (at the point A)

$$\sigma_{R_i} = \sigma_e + \sigma_{bi} = 1 + 34.76 = 35.76 \text{ MPa}$$

Resultant stress at outer surface:

$$\sigma_{R_o} = \sigma_e - \sigma_{bo} = 1 - 23.71 = -22.71 \text{ MPa}$$

∴ Max stress at A = 35.76 MPa ans.



$$15- M = F \cdot x = 10 \times 10^3 \cdot (25 + 25 + 40) = 900 \times 10^3 \text{ N}\cdot\text{mm}$$

$$\frac{M}{I} = \frac{\sigma}{y} \Rightarrow \frac{900 \times 10^3}{\frac{\pi (d)^4}{64}} = \frac{\sigma}{d/2} \Rightarrow \sigma_b = \frac{900 \times 10^3 \times 32}{\pi (75)^3} = 21.73 \text{ MPa}$$

$$T = F \cdot 125 = 10 \times 10^3 \cdot 125 = 1250 \times 10^3 \text{ N}\cdot\text{mm}$$

$$\frac{T}{J} = \frac{\tau}{r} \Rightarrow \frac{1250 \times 10^3}{\frac{\pi (d)^4}{32}} = \frac{\tau}{d/2} \Rightarrow \tau = \frac{16 \times 1250 \times 10^3}{\pi (75)^3} = 15.1 \text{ MPa}$$

$$\sigma_{\max} = \frac{1}{2} \left[\sigma_{t1} + \sqrt{(\sigma_{t1})^2 + 4\tau^2} \right]$$

$$= \frac{1}{2} \left[21.73 + \sqrt{(21.73)^2 + 4(15.1)^2} \right] = 29.46 \text{ MPa} \quad \underline{\text{ans.}}$$

$$\tau_{\max} = \frac{1}{2} \sqrt{(\sigma_{t1})^2 + 4\tau^2} = \frac{1}{2} \sqrt{(21.73)^2 + 4(15.1)^2} = 18.6 \text{ MPa} \quad \underline{\text{ans.}}$$

$$16- \sigma_t = F/A = \frac{4.5 \times 10^3}{50 \times t} = \frac{90}{t} \text{ MPa}$$

$$M = 4.5 \times 10^3 \times 50 = 225 \times 10^3 \text{ N}\cdot\text{mm}$$

$$\frac{M}{I} = \frac{\sigma_b}{y} \Rightarrow \frac{225 \times 10^3}{\frac{bh^3}{12}} = \frac{\sigma}{\frac{h}{2}} \quad b=t, h=50 \text{ mm}$$

$$\sigma_b = \frac{225 \times 10^3 \times 6}{t(50)^2} = \frac{540}{t}$$

$$\sigma_{t \text{ total}} = \sigma_{t0} + \sigma_b = \frac{90}{t} + \frac{540}{t} = \frac{630}{t}$$

$$70 = \frac{630}{t} \Rightarrow t = 9 \text{ mm} \quad \underline{\text{ans.}}$$

17- $P = 5 \text{ kN}$, $\sigma = 28 \text{ MPa}$

$$b = 3t$$

$$P_H = P \sin 60 = 5 \sin 60 = 4.33 \text{ kN}$$

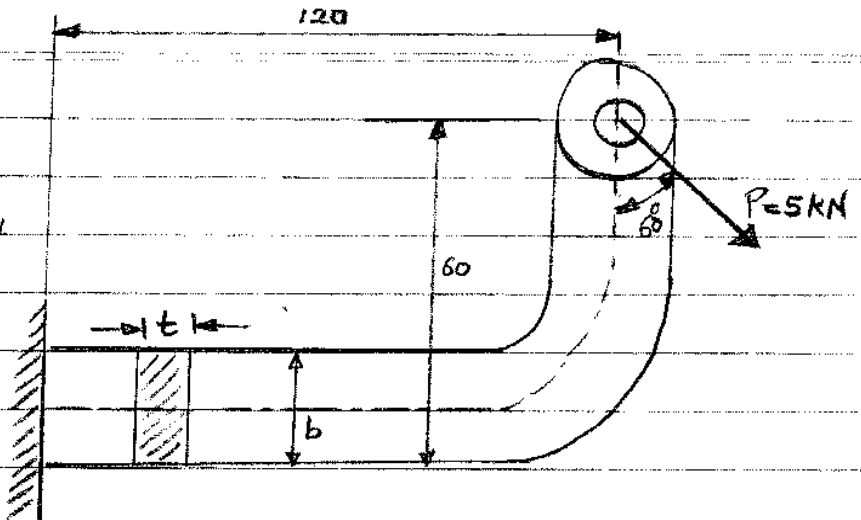
$$P_V = P \cos 60 = 5 \cos 60 = 2.5 \text{ kN}$$

$$M_H = P_H \times 60$$

$$= 4.33 \times 60 = 259.8 \text{ kN}\cdot\text{mm}$$

$$M_V = P_V \times 120$$

$$= 2.5 \times 120 = 300 \text{ kN}\cdot\text{mm}$$



$$\sigma_{bH} = \frac{M_H}{I} \times y = \frac{259.8}{\left(\frac{b t^3}{12}\right)} \times \frac{b}{2} = \frac{259.8 \times 3t \times 10^3}{\frac{t(3t)^3}{12} \times 2} = \frac{173.2 \times 10^3}{t^3}$$

$$\sigma_{bV} = \frac{M_V}{I} \times y = \frac{300 \times 10^3}{\frac{t(3t)^3}{12}} \times \frac{3t}{2} = \frac{200 \times 10^3}{t^3}$$

$$\sigma_{oV} = \frac{P_V}{A} = \frac{2.5 \times 10^3}{t(3t)} = \frac{0.833 \times 10^3}{t^2}$$

$$\sigma_{Total} = \sigma_{bH} + \sigma_{bV} + \sigma_{oV}$$

$$28 = \frac{173.2 \times 10^3}{t^3} + \frac{200 \times 10^3}{t^3} + \frac{0.833 \times 10^3}{t^2}$$

By using try and error

$$t = 24.5 \sim 25 \text{ mm}, b = 3t = 75 \text{ mm}$$

Another Solution : $A = b \cdot t = 3t \cdot t = 3t^2$, $Z = \frac{t \cdot b^2}{6} = \frac{t \cdot (3t)^2}{6} = 1.5t^3$

$$P_H = 5000 \cos 30 = 4330.127 \text{ N}, M_H = 4330.127 \times 60 = 259807.62 \text{ N}\cdot\text{mm}$$

$$\sigma_{bH} = \frac{M_H}{Z} = \frac{259807.62}{1.5t^3} = \frac{173205.08}{t^3}$$

$$P_V = 5000 \sin 30 = 2500 \text{ N}, \sigma_{oV} = P_V / A = 2500 / 3t^2 = \frac{833.33}{t^2}$$

$$M_V = 2500 \times 120 = 300000 \text{ N}\cdot\text{mm}, \sigma_{bV} = \frac{M_V}{Z} = \frac{300000}{1.5t^3} = \frac{200000}{t^3}$$

$$\sigma_{Total} = \sigma_{bV} + \sigma_{bH} + \sigma_{oV}$$

$$28 = \frac{200000}{t^3} + \frac{173205.08}{t^3} + \frac{833.33}{t^2}, \text{ By using try and error method}$$

$$t = 25 \text{ mm}, b = 3t = 75 \text{ mm} \quad (16)$$

$$18- P_H = P \sin 30 = 0.5P$$

$$P_V = P \cos 30 = 0.866P$$

$$M_H = P_H * 50 = 0.5P * 50 = 25P$$

$$M_V = P_V * 160 = 0.866P * 160 = 138.5P$$

$$\sigma_{bH} = \frac{M_H}{I_{xx}} * y = \frac{25P * 12}{30(60)^3} * \frac{60}{2} = 1.389 * 10^{-3} P$$

$$\sigma_{bV} = \frac{M_V}{I_{yy}} * y = \frac{138.5P * 12}{60(30)^3} * \frac{60}{2} = 30.77 * 10^{-3} P$$

$$\sigma_{oV} = P_V / A = 0.866P / 1800 = 0.4811P$$

$$\sigma_{Total} = \sigma_{oV} + \sigma_{bH} + \sigma_{bV}$$

$$70 = P(0.4811 + 1.389 + 30.77) * 10^{-3}$$

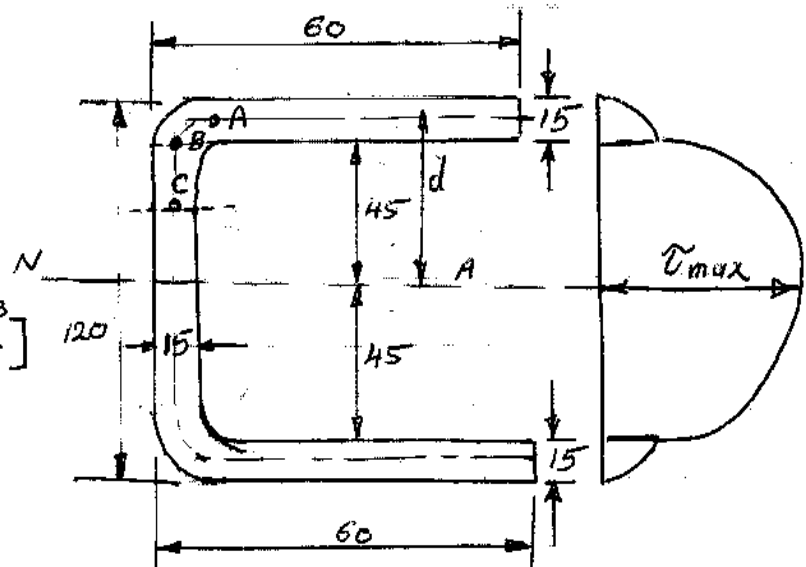
$$P = 2422 \text{ N}$$

20- Symetric section

$$C.A = N.A$$

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = 60$$

$$I = 2 \left[\frac{bh^3}{12} + A \cdot d^2 \right] + \left[\frac{bh^3}{12} \right]$$



$$I = 2 \left[\frac{60(15)^3}{12} + 60 \times 15 \times (45 + 7.5)^2 \right] + \left[\frac{15 \times 90^3}{12} \right] = 5.91 \times 10^6 \text{ mm}^4$$

shear stress:

At point A:

$$\tau_A = \frac{F \cdot A \cdot \bar{y}}{I \cdot b} = \frac{50 \times 10^3 \times 60 \times 15 \times (45 + 7.5)}{5.91 \times 10^6 \times 60} = 6.67 \text{ MPa}$$

At point B:

$$\tau_B = \frac{F \cdot A \cdot \bar{y}}{I \cdot b} = \frac{50 \times 10^3 \times 60 \times 15 \times (45 + 7.5)}{5.91 \times 10^6 \times 15} = 26.7 \text{ MPa}$$

At point C:

$$A_{\text{Total}} = (60 \times 15) + (45 \times 15) = 1575 \text{ mm}^2$$

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{(60 \times 15)(45 + 7.5) + (45 \times 15) \times 22.5}{1575} = 39.6 \text{ mm}$$

$$\tau_c = \tau_{\text{max}}$$

$$\tau_c = \frac{F \cdot A \cdot \bar{y}}{I \cdot b} = \frac{50 \times 10^3 \times 1575 \times 39.6}{5.91 \times 10^6 \times 15} = 35.3 \text{ MPa}$$

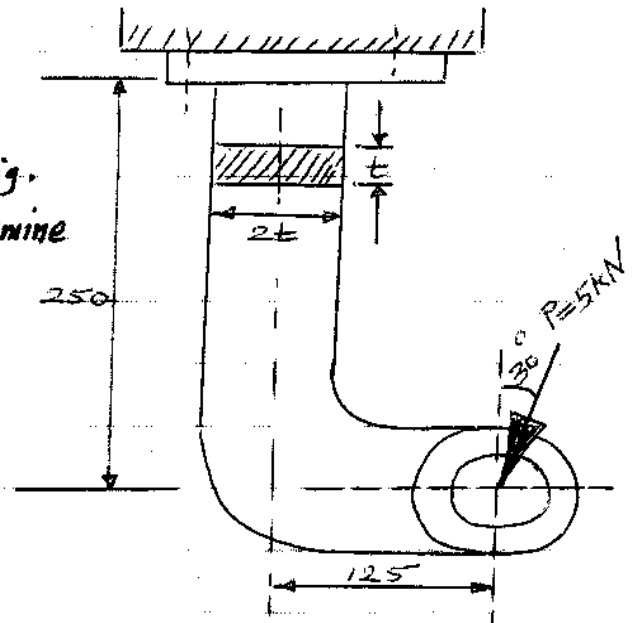
$$\tau_{\text{max}} = F/A$$

$$= \frac{50 \times 10^3}{2(60 \times 15) + (90 \times 15)} = \frac{50 \times 10^3}{3150} = 15.87 \text{ MPa}$$

$$\% \text{ Ratio} = \frac{\tau_{\text{max}}}{\tau_{\text{mean}}} = \frac{35.3}{15.87} = 2.22 \quad \text{ans.}$$

Ex: Quiz

A bracket, made of steel FeE200 ($\sigma_{yt} = 200 \text{ N/mm}^2$) and subjected to a force of 5 kN acting at angle of 30° to the vertical as shown in fig. The factor of safety is 4. Determine the dimensions of the cross-section of the bracket.

Solution:

$$\sigma_y = 200 \text{ N/mm}^2, F.S = 4$$

$$A = 2t \times t = 2t^2 \text{ mm}^2$$

$$F_H = 5000 \sin 30 = 2500 \text{ N}$$

$$F_V = 5000 \cos 30 = 4330 \text{ N}$$

$$M_H = F_H \times 250 = 2500 \times 250 = 625000 \text{ N}\cdot\text{mm}$$

$$M_V = F_V \times 125 = 4330 \times 125 = 541250 \text{ N}\cdot\text{mm}$$

$$\sigma_{ov} = \frac{F_V}{A} = \frac{4330}{2t^2} = \frac{2165}{t^2}$$

$$\sigma_{bH} = \frac{M_H \cdot \bar{Y}}{I} = \frac{625000}{\frac{2t \times t^3}{12}} \times \frac{t}{2} = \frac{1875000}{t^3}$$

$$\sigma_{bV} = \frac{M_V \cdot \bar{Y}}{I} = \frac{541250}{\frac{2t \times t^3}{12}} \times \frac{t}{2} = \frac{1623750}{t^3}$$

$$\frac{\sigma_y}{F.S} = \sigma_{ov} + \sigma_{bH} + \sigma_{bV}$$

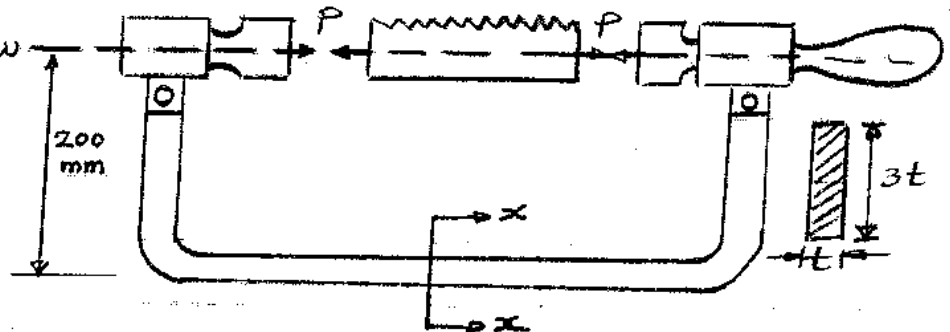
$$\frac{200}{4} = \frac{2165}{t^2} + \frac{1875000}{t^3} + \frac{1623750}{t^3}$$

By solving the final equation mathematically (hit and trail)

$$t = \quad \text{mm}$$

Ex: Quiz

The frame of a hacksaw shown in fig., the initial tension (P) in the blade should be (300N), the frame is made of plain carbon



steel 30C8 with tensile yield strength of 400 N/mm^2 and the factor of safety is 2.5, the cross-section of the frame is rectangular with a ratio of depth to width as 3. Determine the dimensions of the cross-section.

Solution

$$A = 3t^2, \quad \sigma_o = P/A = 300/3t^2 = 100/t^2$$

$$\sigma_b = M/Z = (300 \times 200) / [t(3t)^2/6] = 4 \times 10^4 / t^3$$

$$\sigma_{\text{total}} = \sigma_o + \sigma_b = 100/t^2 + 4 \times 10^4 / t^3 \approx \frac{400}{2.5} = 160$$

$$t = 6.3 \text{ mm}$$

Trail and Error Method

$$At^3 - Bt - C = 0$$

$$t = u + v$$

$$u = \sqrt[3]{-\frac{C}{2} + \sqrt{\frac{C^2}{4} + \frac{B^3}{27}}}$$

$$v = \sqrt[3]{-\frac{C}{2} - \sqrt{\frac{C^2}{4} + \frac{B^3}{27}}}$$

Ex:

Estimate the minimum shaft diameter according to the maximum shear stress theory for a shaft loaded with bending moment of $M = 50 \text{ N.m}$ and a torque of 40 N.m , the material has a yield strength of 1100 MPa .

Solution:

$$M = 50 \text{ N.m}, T = 40 \text{ N.m}, \sigma_y = 1100 \text{ MPa}$$

$$\sigma_b = M/I \cdot y, y = d/2, I = \pi/64 \cdot d^4$$

$$\sigma_b = \frac{M \cdot 64}{\pi d^4} \cdot \frac{d}{2} = \frac{32M}{\pi d^3} = \frac{32 \cdot 50}{\pi d^3} = \frac{509.296}{d^3}$$

$$\tau = \frac{T}{J} \cdot r, r = d/2, J = \frac{\pi}{32} d^4$$

$$\tau = \frac{16T}{\pi d^3} = \frac{16 \cdot 40}{\pi d^3} = \frac{203.716}{d^3}$$

For Maximum shear stress theory:

$$\begin{aligned} \tau_{\max} &= \sqrt{(\sigma_b)^2 + 4\tau^2} \\ &= \sqrt{\left(\frac{509.296}{d^3}\right)^2 + 4\left(\frac{203.716}{d^3}\right)^2} = \frac{326.108}{d^3} \end{aligned}$$

$$\text{F.S.} = \frac{\sigma_y}{2\tau_{\max}} \quad \text{F.S.} = 1 \text{ for minimum shaft dia.}$$

$$\sigma_y = 2\tau_{\max} \Rightarrow 1100 \cdot 10^6 = 2 \cdot \frac{326.108}{d^3}$$

$$d = 8.4 \text{ mm} \simeq 8.5 \text{ mm}$$

Solutions of Problems - Chapter 6

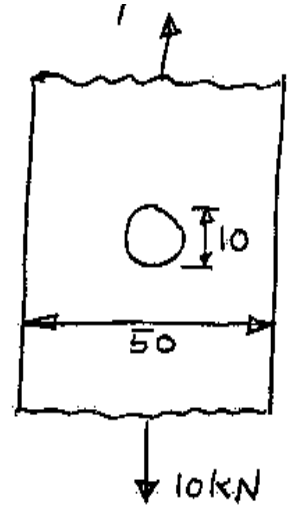
1- Given: $b = 50 \text{ mm}$, $h = 10 \text{ mm}$, $d = 10 \text{ mm}$
 $P = 10 \text{ kN}$

$$\frac{d}{b} = \frac{10}{50} = 0.2$$

from table (6.1) we get: $k_t = 2.5$

$$\sigma_{\text{normal}} = P/A = \frac{10 \times 10^3}{50 \times 10} = 20 \text{ N/mm}^2$$

$$\sigma_{\text{max}} = \sigma_{\text{normal}} * k_t = 20 * 2.5 = 50 \text{ MPa}$$



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2- Given: $D = 45 \text{ mm}$, $d = 30 \text{ mm}$, $r = 6 \text{ mm}$, $P = 10 \text{ kN}$

$$D/d = 45/30 = 1.5 \text{ , } r/d = 6/30 = 0.2$$

from table (6.3) we get: $k_t = 1.56$

$$\sigma_o = P/A = 10 \times 10^3 / (\pi/4 d^2) = \frac{10 \times 10^3}{\frac{\pi}{4} (30)^2} = 14.154 \text{ MPa}$$

$$\sigma_{\text{max}} = \sigma_o * k_t = 14.154 * 1.56 = 22 \text{ MPa}$$

—————x—————x—————x—————x—————x—————x—————x—————x—————x—————x—————

$$5- F_{mean} = \frac{F_{max} + F_{min}}{2} = \frac{10 + (-5)}{2} = \frac{10-5}{2} = 2.5 \text{ kN}$$

$$F_{variable} = \frac{F_{max} - F_{min}}{2} = \frac{10 - (-5)}{2} = \frac{15}{2} = 7.5 \text{ kN}$$

$$\sigma_m = \frac{F_m}{A} = \frac{2.5 \times 10^3}{\frac{\pi}{4}(d)^2} = \frac{10 \times 10^3}{\pi d^2}$$

$$\sigma_v = \frac{F_v}{A} = \frac{7.5 \times 10^3}{\frac{\pi}{4}(d)^2} = \frac{30 \times 10^3}{\pi d^2}$$

choose factors:

$$q=1, K_f = K_t = 2.2,$$

from fig. 6.3 according to $\sigma_u = 600 \text{ MPa}$ we get:

$$K_{sur} = 0.75, K_a = 0.8, K_{sz} = 0.85 \text{ for } d < 50 \text{ mm}$$

Then: By using Soderberg Method:

$$\frac{1}{F.S} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v * K_f}{\sigma_e * K_a * K_b * K_{sur} * K_{sz} * K_i}$$

$$\frac{1}{4} = \frac{\frac{10 \times 10^3}{\pi d^2}}{380} + \frac{\frac{30 \times 10^3}{\pi d^2} * 2.2}{290 * 0.8 * 1 * 0.75 * 0.85 * 1}$$

$$d = 24.53 \text{ mm} \quad \text{ans.}$$

$$6 - F_{max} = 25 \text{ kN}, F_{min} = -15 \text{ kN}, K_{sur} = 0.88, K_f = 2.25$$

$$\sigma_e = 360 \text{ MPa}, \sigma_y = 400 \text{ MPa}, K_i = 0.25, F.S = 1.5$$

Assume $K_{sz} = 0.89$

$$F_m = \frac{F_{max} + F_{min}}{2} = \frac{25 - 15}{2} = 5 \text{ kN}, \sigma_m = \frac{F_m}{A} = \frac{5 \times 10^3 * 4}{\pi d^2} = \frac{20 \times 10^3}{\pi d^2}$$

$$F_v = \frac{F_{max} - F_{min}}{2} = \frac{25 + 15}{2} = 20 \text{ kN}, \sigma_v = \frac{F_v}{A} = \frac{20 \times 10^3 * 4}{\pi d^2} = \frac{80 \times 10^3}{\pi d^2}$$

By using Soderberg Method:

$$\frac{1}{F.S} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v * K_f}{\sigma_e * K_i * K_{sz} * K_{sur}}$$

$$\frac{1}{1.5} = \frac{\frac{20 \times 10^3}{\pi d^2}}{400} + \frac{\frac{80 \times 10^3}{\pi d^2}}{360 * 0.25 * 0.88 * 0.89}$$

$$d = 35.25 \approx 35.3 \text{ mm} \quad \text{ans.} \quad (23)$$

7-

$F_{max} = 160 \text{ kN}$, $F_{min} = -160 \text{ kN}$ since the load is axial completely reversed

So $F_{mean} = 0$ then $\sigma_m = 0$

Assume $k_a = 0.8$, $\sigma_e = 0.5 \sigma_u = 0.5 \times 1100 = 550 \text{ MPa}$

$$F_v = \frac{160 - (-160)}{2} = \frac{320}{2} = 160 \text{ kN}$$

$$\sigma_v = \frac{F_v}{A} = \frac{160 \times 10^3}{\frac{\pi (d)^2}{4}} = \frac{640 \times 10^3}{\pi d^2}$$

By using Soderberg Method:

$$\frac{1}{FIS} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v \cdot k_f}{\sigma_e \cdot k_a \cdot k_{sz} \cdot k_{sur} \cdot k_i}$$

$$\frac{1}{2} = \frac{0}{930} + \frac{\frac{640 \times 10^3}{\pi d^2} \cdot 1}{550 \cdot 0.8 \cdot 1 \cdot 1 \cdot 1}$$

$$d = 30.43 \text{ mm} \quad \text{ans)}$$

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8- $F_{max} = 200 \text{ kN}$, $F_{min} = -200 \text{ kN}$ since the load is axial completely reversed

So $F_{mean} = 0$ then $\sigma_m = 0$, $\sigma_y = 440 \text{ MPa}$, $\sigma_u = 600 \text{ MPa}$

$k_{sur} = 0.8$, $FIS = 1.5$, $k_a = 0.8$, $k_f = 1$, $k_{sz} = 0.75$

$$\sigma_e = 0.5 \sigma_u = 0.5 \times 600 = 300 \text{ MPa}, F_v = \frac{200 + 200}{2} = 200 \text{ kN}, \sigma_v = \frac{800 \times 10^3}{\pi d^2}$$

By using Soderberg Method:

$$\frac{1}{FIS} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v \cdot k_f}{\sigma_e \cdot k_a \cdot k_{sur} \cdot k_{sz} \cdot k_b}$$

$$\frac{1}{1.5} = \frac{0}{440} + \frac{\frac{800 \times 10^3}{\pi d^2} \cdot 1}{300 \cdot 0.8 \cdot 0.8 \cdot 0.75 \cdot 1}$$

$$d = 51.5 \text{ mm} \quad \text{ans)}$$

9- Given: $T_{max} = 1600 \text{ N.m}$, $T_{min} = 800 \text{ N.m}$

$$\sigma_u = 600 \text{ MPa}, \sigma_y = 450 \text{ MPa}, k_{fs} = 1.2, k_{sur} = 0.8, k_{sz} = 0.85$$

$$\text{Let } F_s = 1, \sigma_e = 0.5 \sigma_u = 0.5 * 600 = 300 \text{ MPa}$$

$$\tau_e = 0.55 \sigma_e = 0.55 * 300 = 165 \text{ MPa}$$

$$\tau_y = 0.5 \sigma_y = 0.5 * 450 = 225 \text{ MPa}$$

$$T_m = \frac{1600 + 800}{2} = 1200 * 10^3 \text{ N.mm}, T_v = \frac{1600 - 800}{2} = 400 * 10^3 \text{ N.mm}$$

$$\tau_m = \frac{T_m}{J} \cdot r = \frac{1200 * 10^3 * 16}{\pi d^3}, \tau_v = \frac{T_v}{J} \cdot r = \frac{400 * 10^3 * 16}{\pi d^3}$$

According to Soderberg's formula:

$$\frac{1}{F_s} = \frac{\tau_m}{\tau_y} + \frac{\tau_v * k_{fs}}{\tau_e * k_{sur} * k_{sz}}$$

$$\frac{1}{1} = \frac{\frac{1200 * 10^3 * 16}{\pi d^3}}{225} + \frac{\frac{400 * 10^3 * 16}{\pi d^3} * 1.2}{165 * 0.8 * 0.85}$$

$$d = 36.58 \text{ mm} \quad \text{ans)}$$

10- Given: $\sigma_e = 600 \text{ MPa}$, $k_{sur} = 0.87$, $k_{sz} = 0.85$

$$F_s = 1.6, F_{max} = 10 \text{ kN}, F_{min} = -10 \text{ kN}, k_f = 1$$

$$M_v = \frac{W \cdot l}{4} = \frac{10 * 450 * 10^3}{4} = 1125 * 10^3 \text{ N.mm}$$

$$Z = \frac{\pi}{32} d^3$$

$$\sigma_m = \frac{M_v}{Z} = \frac{1125 * 10^3}{\frac{\pi}{32} d^3} = \frac{36 * 10^6}{\pi d^3}$$

steady load $\sigma_m = 0$

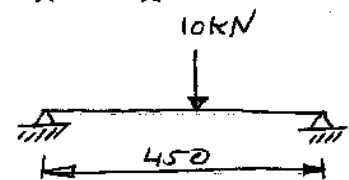
By using Soderberg's formula:

$$\frac{1}{F_s} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v * k_f}{\sigma_e * k_{sur} * k_{sz}}$$

$$\frac{1}{1.6} = \frac{0}{\sigma_y} + \frac{\frac{36 * 10^6}{\pi d^3} * 1}{600 * 0.87 * 0.85}$$

$$d = 34.57 \approx 35 \text{ mm} \quad \text{ans)}$$

(25)



11- $\sigma_e = 280 \text{ MPa}$, $\sigma_y = 350 \text{ MPa}$, $k_t = 1.8$, $F.S = 2$

$F_{\max} = 700 \text{ kN}$, $F_{\min} = -300 \text{ kN}$

Let $q = 0$, So $k_f = 1 + q(k_t - 1) = 1$

$F_m = \frac{F_{\max} + F_{\min}}{2} = \frac{700 + (-300)}{2} = 200 \text{ kN}$

$F_v = \frac{F_{\max} - F_{\min}}{2} = \frac{700 - (-300)}{2} = 500 \text{ kN}$

$\sigma_m = \frac{F_m}{A} = \frac{200 \times 10^3}{\frac{\pi}{4} d^2} = \frac{800 \times 10^3}{\pi d^2}$

$\sigma_v = \frac{F_v}{A} = \frac{500 \times 10^3}{\frac{\pi}{4} d^2} = \frac{2000 \times 10^3}{\pi d^2}$

By using Soderberg's formula :

$$\frac{1}{F.S} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v * k_f}{\sigma_e * k_a * k_b * k_{sur} * k_{sz}}$$

$$\frac{1}{2} = \frac{\frac{800 \times 10^3}{\pi d^2}}{350} + \frac{\frac{2000 \times 10^3}{\pi d^2} * 1}{280 * 1 * 1 * 1 * 1}$$

$d = 77.49 \simeq 80 \text{ mm}$

12- Given: Cold drawn steel rod

$$F_{b \max} = 1130, F_{b \min} = 565, F_{a \max} = 13500 \text{ N}, F_{a \min} = 4500 \text{ N}$$

$$k_{sz} = 0.85, k_{sur} = 0.89, k_b = 1, k_a = 0.7, k_f = 1, F.S = 2$$

$$\sigma_u = 550 \text{ MPa}, \sigma_y = 470 \text{ MPa}, \sigma_e = 0.5 \sigma_u = 0.5 \times 550 = 275 \text{ MPa}$$

Solution: For bending Load:

$$M_m = \frac{M_{\max} + M_{\min}}{2} = \frac{1130 + 565}{2} = 847.5 \times 10^3 \text{ N}\cdot\text{mm}$$

$$M_v = \frac{M_{\max} - M_{\min}}{2} = \frac{1130 - 565}{2} = 282.5 \times 10^3 \text{ N}\cdot\text{mm}$$

$$\sigma_{mb} = \frac{M_m}{Z} = \frac{847.5 \times 10^3 \times 32}{\pi d^3}, \sigma_{mv} = \frac{M_v}{Z} = \frac{282.5 \times 10^3 \times 32}{\pi d^3}$$

$$\sigma_{ne_b} = \sigma_m + \frac{\sigma_v \times \sigma_y \times k_f}{\sigma_e \times k_b \times k_{sur} \times k_{sz}}$$

$$= \frac{847.5 \times 10^3 \times 32}{\pi d^3} + \frac{282.5 \times 10^3 \times 32 \times 470 \times 1}{275 \times 1 \times 0.89 \times 0.85} = \frac{15133.48 \times 10^3}{d^3}$$

For axial Load:

$$F_m = \frac{F_{\max} + F_{\min}}{2} = \frac{13500 + 4500}{2} = 9000 \text{ N}$$

$$F_v = \frac{F_{\max} - F_{\min}}{2} = \frac{13500 - 4500}{2} = 4500 \text{ N}$$

$$\sigma_m = \frac{F_m}{A} = \frac{9000 \times 4}{\pi d^2}, \sigma_v = \frac{F_v}{A} = \frac{4500 \times 4}{\pi d^2}$$

$$\sigma_{ne_d} = \sigma_m + \frac{\sigma_v \times \sigma_y \times k_f}{\sigma_e \times k_a \times k_b \times k_{sur} \times k_{sz}}$$

$$= \frac{9000 \times 4}{\pi d^2} + \frac{4500 \times 4 \times 470 \times 1}{275 \times 0.7 \times 1 \times 0.89 \times 0.85} = \frac{29951}{d^2}$$

$$\sigma_{\max} = \sigma_y / F.S$$

$$\sigma_{\max} = \sigma_{ne} = \sigma_{ne_b} + \sigma_{ne_d} = \frac{15133.48 \times 10^3}{d^3} + \frac{29951}{d^2} = \frac{470}{2}$$

By using hit and trail:

$$d = 40.5 \approx 41 \text{ mm}$$

13- Given: $\sigma_u = 550 \text{ MPa}$, $\sigma_y = 470 \text{ MPa}$, F.S = 2, $k_{fb} = 1.44$, $k_{sz} = 0.85$

$$\sigma_e = 275 \text{ MPa}, k_{fa} = 1.63$$

For axial Load:

$$W_m = \frac{W_{\max} + W_{\min}}{2} = \frac{450 + (-110)}{2} = 170 \text{ N}$$

$$W_v = \frac{W_{\max} - W_{\min}}{2} = \frac{450 - (-110)}{2} = 280 \text{ N}$$

$$\sigma_m = \frac{W_m}{A} = \frac{170 \times 4}{\pi d^2} = \frac{680}{\pi d^2}, \quad \sigma_v = \frac{W_v}{A} = \frac{280 \times 4}{\pi d^2} = \frac{1120}{\pi d^2}$$

$$\sigma_{nea} = \sigma_m + \frac{\sigma_v \times \sigma_y \times k_{fa}}{\sigma_e \times k_a \times k_b \times k_{sur} \times k_{sz}}$$

$$= \frac{680}{\pi d^2} + \frac{\frac{1120}{\pi d^2} \times 470 \times 1.63}{275 \times 1 \times 1 \times 0.9 \times 0.85} = \frac{1515.472}{d^2}$$

For bending Load:

$$W_m = \frac{W_{\max} + W_{\min}}{2} = \frac{135 + 45}{2} = 90 \text{ N}, \quad M_m = 90 \times 125 = 11250 \text{ N}\cdot\text{mm}$$

$$W_v = \frac{W_{\max} - W_{\min}}{2} = \frac{135 - 45}{2} = 45 \text{ N}, \quad M_v = 45 \times 125 = 5625 \text{ N}\cdot\text{mm}$$

$$Z = \frac{\pi d^3}{32}$$

$$\sigma_{mb} = \frac{M_m}{Z} = \frac{11250 \times 32}{\pi d^3} = \frac{114649.68}{d^3}$$

$$\sigma_{vb} = \frac{M_v}{Z} = \frac{5625 \times 32}{\pi d^3} = \frac{57324.841}{d^3}$$

$$\sigma_{neb} = \sigma_{mb} + \frac{\sigma_v \times \sigma_y \times k_{fb}}{\sigma_e \times k_a \times k_b \times k_{sur} \times k_{sz}}$$

$$= \frac{114649.68}{d^3} + \frac{\frac{57324.841}{d^3} \times 470 \times 1.44}{275 \times 1 \times 0.9 \times 0.85} = \frac{299070.13}{d^3}$$

$$\sigma_{ne} = \sigma_{nea} + \sigma_{neb} = \sigma_{\max} = \frac{\sigma_y}{F.S}$$

$$\frac{1515.472}{d^2} + \frac{299070.13}{d^3} = \frac{470}{2}$$

By hit and trial:

$$d = 11.035 \approx 11.5$$

ans)

14- Given: $M_v = 800 \text{ N.m}$, $T = 500 \text{ N.m}$, $\pm 40\%$, $k_{fs} = 1.2$, $\sigma_y = 500 \text{ MPa}$

$$\sigma_u = 800 \text{ MPa}, \sigma_e = 400 \text{ MPa}.$$

$$\tau_y = 0.5 \sigma_y = 0.5 * 500 = 250 \text{ MPa}$$

$$T_{\max} = T + (T * 40\%) = 500 + (500 * \frac{40}{100}) = 700 \text{ N.m}$$

$$T_{\min} = T - (T * 40\%) = 500 - (500 * \frac{40}{100}) = 300 \text{ N.m}$$

$$T_m = \frac{T_{\max} + T_{\min}}{2} = \frac{700 + 300}{2} = 500 \text{ N.m}$$

$$T_v = \frac{T_{\max} - T_{\min}}{2} = \frac{700 - 300}{2} = 200 \text{ N.m}$$

$$\tau_m = \frac{16 T_m}{\pi d^3} = \frac{16 * 500 * 10^3}{\pi d^3} = \frac{8 * 10^6}{\pi d^3}$$

$$\tau_v = \frac{16 T_v}{\pi d^3} = \frac{16 * 200 * 10^3}{\pi d^3} = \frac{3.2 * 10^6}{\pi d^3}$$

$$\tau_{\text{nes}} = \tau_m + \frac{\tau_v * \tau_y * k_{fs}}{\sigma_e * k_{sz} * k_{sur}} = \frac{8 * 10^6}{\pi d^3} + \frac{\frac{3.2 * 10^6}{\pi d^3} * 250 * 1.2}{400 * 0.55} = \frac{3.937 * 10^6}{d^3}$$

$$\sigma_m = 0$$

$$\sigma_{\text{neb}} = \sigma_m + \frac{\sigma_u * \sigma_y * k_f}{\sigma_e} = 0 + \frac{800 * 10^3 * 32 * 500}{400} =$$

$$= 0 + \frac{\frac{8 * 10^3}{\pi d^3} * 500 * 1.2}{400} = \frac{3.8217 * 10^6}{d^3}$$

According to Max. shear stress theory:

$$\frac{250}{F.S} = \frac{1}{2} \sqrt{(\sigma_e)^2 + 4 \tau^2}$$

$$\frac{250}{1.2} = \frac{1}{2} \sqrt{\left(\frac{3.8217 * 10^6}{d^3}\right)^2 + 4 \left(\frac{3.937 * 10^6}{d^3}\right)^2}$$

By solving the final equation mathematically:

$$d = 40.8 \approx 41 \text{ mm}$$

ans

15-

$$T_{max} = 165, T_{min} = 55, M_{max} = 510, M_{min} = -170, \sigma_u = 540 \text{ MPa}$$

$$\sigma_y = 400 \text{ MPa}, k_{fb} = 1.6, k_{fs} = 1.3, F.S. = 1.5, K_{sz} = 0.85, K_{sur} = 0.88$$

$$M_m = \frac{M_{max} + M_{min}}{2} = \frac{510 + (-170)}{2} = 170 \text{ N.m}$$

$$M_v = \frac{M_{max} - M_{min}}{2} = \frac{510 - (-170)}{2} = 340 \text{ N.m}$$

$$Z = \frac{\pi}{32} d^3 = 0.0982 d^3, \sigma_m = M_m / Z = 170 / 0.0982 d^3 = 1731.16 / d^3$$

$$\sigma_v = M_v / Z = 340 / 0.0982 d^3 = 3462.321 / d^3$$

$$\text{Assume that } \sigma_e = 0.5 \sigma_u = 540 * 0.5 = 270 \text{ N/mm}^2$$

$$\sigma_{eb} = \sigma_e = \frac{\sigma_u}{2} = \frac{540}{2} = 270$$

$$\sigma_{neb} = \sigma_{ne} = \sigma_m + \frac{\sigma_v * \sigma_y * k_{fb}}{\sigma_{eb} * K_{sur} * K_{sz}}$$

$$= \frac{1731.16}{d^3} + \frac{3462.322}{d^3} * \frac{400 * 1.6}{270 * 0.88 * 0.85} = \frac{12703.066}{d^3}$$

$$T_m = \frac{T_{max} + T_{min}}{2} = \frac{165 + 55}{2} = 110 \text{ N.m}$$

$$T_v = \frac{T_{max} - T_{min}}{2} = \frac{165 - 55}{2} = 55 \text{ N.m}$$

$$\tau_m = 16 T_m / \pi d^3 = 16 * 110 / \pi d^3 = 560.509 / d^3$$

$$\tau_v = 16 T_v / \pi d^3 = 16 * 55 / \pi d^3 = 280.254 / d^3$$

$$\tau_e = 0.55 \sigma_e = 0.55 * 270 = 148.5$$

$$\tau_y = 0.5 \sigma_y = 0.5 * 400 = 200$$

$$\tau_{es} = \tau_m + \frac{\tau_v * \tau_y * k_{fs}}{\tau_e * K_{sur} * K_{sz}}$$

$$= \frac{560.509}{d^3} + \frac{280.254}{d^3} * \frac{200 * 1.3}{148.5 * 0.88 * 0.85} = \frac{1216.498}{d^3}$$

$$\tau_{es(max)} = \frac{\tau_y}{F.S.} = \frac{1}{2} \sqrt{(\sigma_{ne})^2 + 4(\tau_{es})^2}$$

$$\frac{400}{1.5} = \frac{1}{2} \sqrt{\left(\frac{12703.066}{d^3}\right)^2 + 4\left(\frac{1216.498}{d^3}\right)^2}$$

$$d = 35.8 \text{ mm} \approx 36.5 \text{ ans}$$

Solutions of ch. 10

1-

For static load

$$P = 0.707 * S * l * \sigma_E = 2 * 0.707 * 10 * l * 70 \Rightarrow l = 70.72$$

$$L_{Total} = 70.72 + 12.5 = 83.2 \text{ mm} \quad \underline{\text{ans.}}$$

For Fatigue load

$$K_w = 1.5 \quad \text{From table 10.6}$$

$$\sigma_E = 70/1.5$$

$$\therefore P = 2 * 0.707 * 10 * l * \frac{70}{1.5}$$

$$70 * 10^3 = 2 * 0.707 * 10 * l * 70/1.5 \Rightarrow l = 106.8 \text{ mm}$$

$$L_{Total} = 106.8 + 12.5 = 118.58$$

—————x—————x—————x—————x—————x—————x—————x—————

2-

For static load

$$P = 2 * 0.707 * S * l * \sigma$$

$$70 * 10^3 = 2 * 0.707 * 10 * l * 56 \Rightarrow l = 88.4 \text{ mm}$$

$$L_{Total} = 88.4 + 12.5 = 100.9 \text{ mm}$$

For Fatigue load

$$K_w = 2.7 \quad (\text{for parallel} - \text{from table 10.6})$$

$$\sigma = 56/2.7$$

$$P = 2 * 0.707 * S * l * \sigma$$

$$70 * 10^3 = 2 * 0.707 * 10 * l * 56/2.7 \Rightarrow l = 238.68 \text{ mm}$$

adding 12.5 mm for starting and stopping weld run

$$L_{Total} = 238.68 + 12.5 = 251.2 \text{ mm}$$

$$3- \quad L = \frac{\text{Tensile load}}{\text{load per mm}} = \frac{180 \times 10^3}{430} = 418.6 \text{ mm}$$

$$b = \bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{85 \times 10 \times 5 + 125 \times 10 \times 62.5}{125 \times 10 + 85 \times 10} = 39.226 \text{ mm}$$

$$d = 125 - 39.226 = 85.774 \text{ mm} \quad (a+b=125)$$

$$L_a = \frac{L \times b}{a+b} = \frac{418.6 \times 39.226}{125} = 131.36 \text{ mm}$$

$$L_b = \frac{L \times a}{a+b} = \frac{418.6 \times 85.774}{125} = 287.24 \text{ mm}$$

checking:

$$L_a + L_b = 131.36 + 287.24 = 418.6 \text{ mm}$$

Note: The adges of 125 mm leg instead of 150 mm leg.

— x — x — x — x — x — x — x — x — x —

4-

$$M = P \times l = 5 \times 10^3 \times 200 = 1000 \times 10^3 \text{ N.mm}$$

$$\sigma_{b \max} = \frac{5.66 M}{\pi \cdot S \cdot d^2}$$

$$100 = \frac{5.66 \times 1000 \times 10^3}{\pi \times S \times (50)^2} \Rightarrow S = 72 \text{ mm} \quad \underline{\text{ans.}}$$

— x — x — x — x — x — x — x — x — x —

$$5- \quad T = 3 \text{ kN.m} = 3 \times 10^3 \times 10^3 = 3 \times 10^6 \text{ N.mm}$$

$$\tau_{\max} = \frac{2.83 T}{\pi \cdot S \cdot d^2}$$

$$70 = \frac{2.83 \times 3 \times 10^6}{\pi \times S \times (65)^2} \Rightarrow S = 9.13 \text{ mm} \approx 10 \text{ mm}$$

if $d = 62 \text{ mm}$ then $S = 10 \text{ mm} \quad \underline{\text{ans.}}$

— x — x — x — x — x — x — x — x — x —

$$6- \quad \tau_{\max} = \frac{4.242 T}{S \cdot L^2} \times 2 \quad \text{for four sides}$$

$$85 = \frac{4.242 \cdot T}{5(8)^2} \times 2 \Rightarrow T = 320.603 \text{ N.m} = 32.06 \text{ kN.m}$$

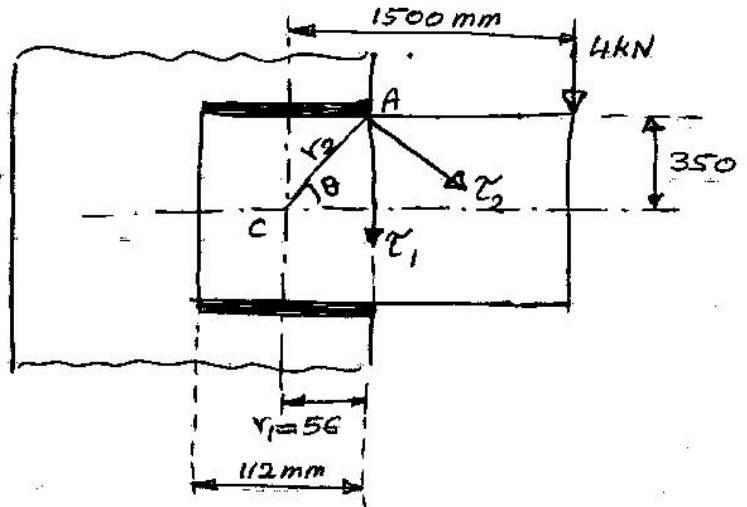
7-

$$r_2 = \sqrt{(56)^2 + (350)^2} = 354.45 \text{ mm}$$

$$J = \frac{t \cdot L \cdot (3b^2 + L^2)}{6}$$

$$= \frac{t \cdot 112 [3(700)^2 + (112)^2]}{6}$$

$$= 27674154.66 t$$



$$\tau_1 = P/A = \frac{4 \times 10^3}{112 \times 2 \times t} = \frac{17.857}{t} \text{ MPa}$$

$$\tau_2 = \frac{P \cdot e \cdot r_2}{J} = \frac{4 \times 10^3 + 1500 \times 354.45}{27674154.66 t} = \frac{76.85}{t} \text{ MPa}$$

$$\cos \theta = \frac{r_1}{r_2} = \frac{56}{354.452} = 0.158$$

$$\tau = \frac{1}{t} \left[(17.857)^2 + (76.85)^2 + 2 \times 17.857 \times 76.85 \times 0.158 \right]^{1/2}$$

$$= \frac{81.6}{t} = \tau_{\text{weld}} = 60 \Rightarrow t = 1.36 \Rightarrow s = \frac{1.36}{0.707} = 1.923 \text{ mm}$$

∴ thickness = 1.923 ≈ 2 mm ans

checking for plate:

$$\sigma_b = \frac{M}{I} \cdot y = \frac{4000 \times 1500}{\frac{2 \times (700)^3}{12}} \times \frac{700}{2} = 36.73 \text{ MPa}$$

$$\tau = P/A = 4 \times 10^3 / 700 \times 2 = 2.86 \text{ MPa}$$

$$\tau_{\text{max}} = \frac{1}{2} \sqrt{(36.73)^2 + 4(2.86)^2} = 18.58 \text{ MPa}$$

Since $\tau_{\text{max}} < 40 \text{ MPa}$ (allowable stress of the plate) Then
our design in the safe region.

8-

$$r_2 = \sqrt{(r_1)^2 + (AB)^2}$$

$$= \sqrt{(50)^2 + (62.5)^2}$$

$$= 80.04 \text{ mm}$$

$$J = \frac{t \cdot L (3b^2 + L^2)}{6}$$

$$= \frac{0.707 \times 10 \times 100 [3(125)^2 + (100)^2]}{6}$$

$$= 6701770.833 \text{ mm}^4$$

Secondary shear stress:

$$\tau_2 = \frac{P \cdot e \cdot r_2}{J} = \frac{16 \times 10^3 \times 300 \times 80.04}{6701770.833} = 57.33 \text{ MPa}$$

Direct shear stress:

$$\tau_1 = P/A = \frac{16 \times 10^3}{2 \times 0.707 \times 10 \times 100} = 11.315 \text{ MPa}$$

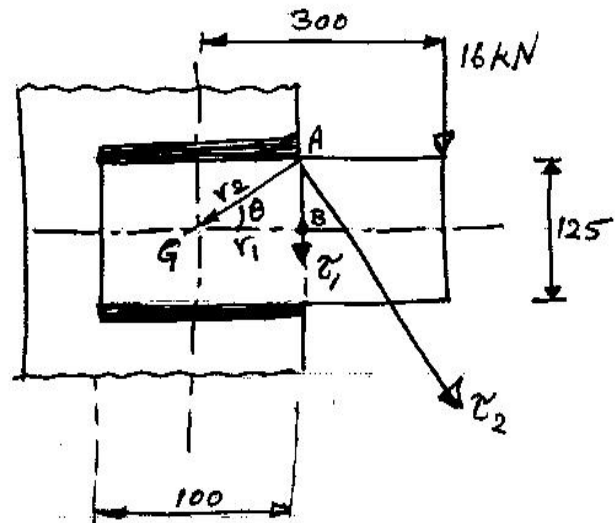
$$\cos \theta = r_1/r_2 = 50/80.04 = 0.6246$$

$$\tau = [\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta]^{1/2}$$

$$= [(11.315)^2 + (57.33)^2 + 2 \times 11.315 \times 57.33 \times 0.6246]^{1/2}$$

$$= 65 \text{ MPa}$$

$$\text{if } t = 10 \text{ mm then } \tau = 45.9 \text{ MPa}$$



9-

$$J = \frac{t \cdot L [3b^2 + L^2]}{6} = \frac{t \cdot 60 [3(100)^2 + (60)^2]}{6} = 336000t \text{ mm}^4$$

$$r_2 = \sqrt{(r_1)^2 + (AB)^2}$$

$$= [(30)^2 + (50)^2]^{1/2}$$

$$= 58.31 \text{ mm}$$

$$\cos \theta = r_1 / r_2 = 30 / 58.31$$

$$= 0.514$$

Primary (direct) shear stress

$$\tau = P/A = \frac{10 \times 10^3}{2 \times t \times 60} = \frac{83.333}{t} \text{ MPa}$$

Secondary shear stress:

$$\tau_2 = \frac{P \cdot e \cdot r_2}{J} = \frac{10 \times 10^3 \times 150 \times 58.31}{336000t} = \frac{260.3}{t} \text{ MPa}$$

$$\tau_{max} = [\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta]^{1/2}$$

$$= \left[\left(\frac{83.33}{t} \right)^2 + \left(\frac{260.3}{t} \right)^2 + 2 \times \frac{83.33}{t} \times \frac{260.3}{t} \times 0.514 \right]^{1/2}$$

$$80 = \frac{1}{t} \times 311.46$$

$$t = 3.893 \text{ mm}$$

$$S = \frac{t}{0.707} = \frac{3.893}{0.707} = 5.51 \text{ mm}$$

10-

$$\bar{x} = \frac{L^2}{2(L+b)} = \frac{(100)^2}{2(100+150)}$$

$$= 20 \text{ mm}$$

$$\bar{y} = \frac{b^2}{2(L+b)} = \frac{(150)^2}{2(100+150)}$$

$$= 45 \text{ mm}$$

$$J = t \left[\frac{(b+t)^3}{12(L+b)} - 6 \times b^2 \times L^2 \right]$$

$$= 0.7075 \left[\frac{(150+100)^3}{12(100+150)} - 6(150)^2(100)^2 \right]$$

$$= 602422.916 \text{ S}$$

$$r_2 = \sqrt{(r_1)^2 + (AB)^2}$$

$$= [(100-20)^2 + (45)^2]^{1/2}$$

$$= 91.788 \text{ mm}$$

$$\cos \theta = r_1 / r_2 = 80 / 91.788 = 0.87$$

Primary (direct) shear stress

$$\tau_1 = P/A = 20 \times 10^3 / (100 \times 0.7075 + 150 \times 0.7075) = \frac{113.154}{S} \text{ MPa}$$

Secondary shear stress

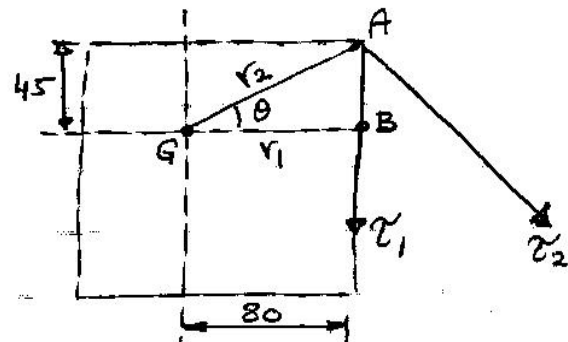
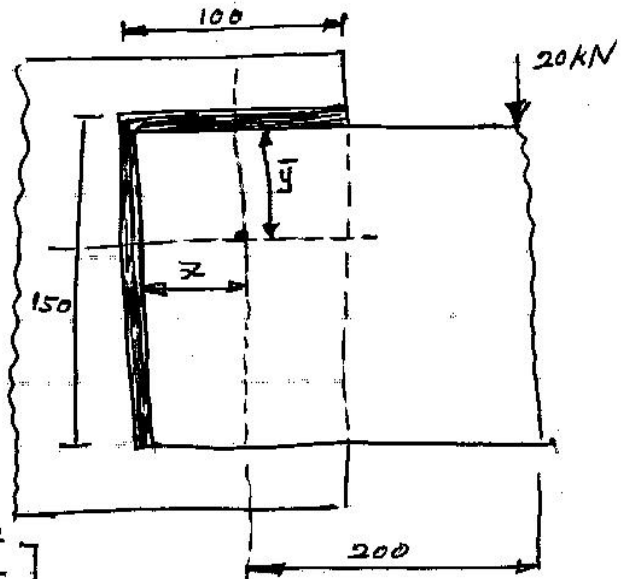
$$\tau_2 = \frac{P \cdot e \cdot r_2}{J} = \frac{20 \times 10^3 \times 200 \times 91.788}{602422.916 S} = \frac{609.46}{S} \text{ MPa}$$

$$\tau_{\max} = [\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta]^{1/2}$$

$$= \left[\left(\frac{113.154}{S} \right)^2 + \left(\frac{609.46}{S} \right)^2 + 2 \times \frac{113.154}{S} \times \frac{609.46}{S} \times 0.87 \right]^{1/2}$$

$$= \frac{710}{S} \text{ MPa}$$

$$80 = \frac{710}{S} \Rightarrow S = 8.875 \approx 8.9 \text{ mm (ans)}$$



11-

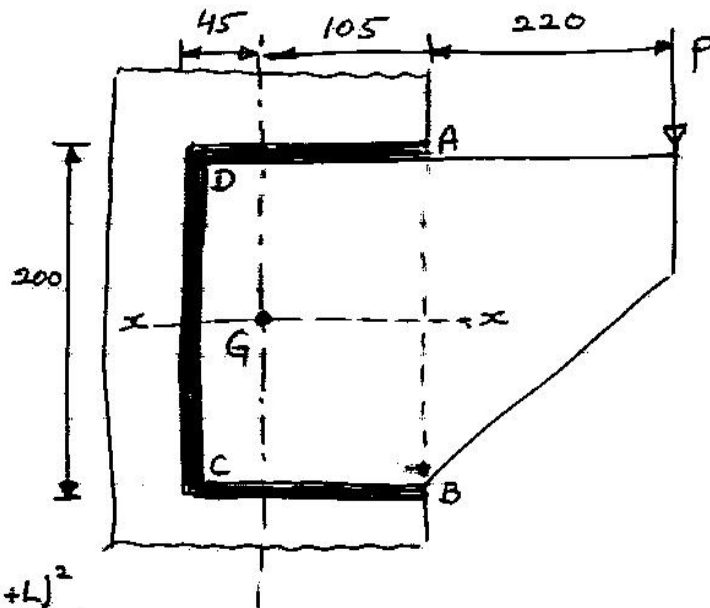
$$x = \frac{L^2}{2L + b}$$

$$= \frac{(150)^2}{2 \times 150 + 200}$$

$$= 45$$

$$r_2 = [(150 - 45)^2 + (100)^2]^{1/2}$$

$$= 145 \text{ mm}$$



$$J = t \left[\frac{(b+2L)^3}{12} - \frac{(L^2(b+L)^2}{b+2L} \right]$$

$$\cos \theta = r_1 / r_2 = 105 / 145 = 0.724$$

$$= 0.707 \times 10 \left[\frac{(200 + 2(150))^3}{12} - \frac{(150)^2(200 + 150)}{200 + 2(150)} \right]$$

$$= 34672458.32 \text{ mm}^4$$

Primary (direct) shear stress:

$$\tau_1 = P / A$$

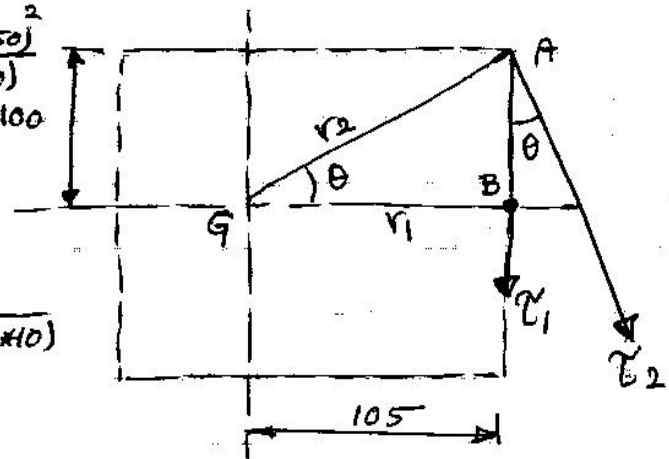
$$= \frac{P}{(2 \times 150 \times 0.707 \times 10) + (200 \times 0.707 \times 10)}$$

$$= 2.828 \times 10^{-4} P \text{ MPa}$$

Secondary shear stress:

$$\tau_2 = \frac{P \cdot e \cdot r_2}{J}$$

$$= \frac{P \times (220 + 105) \times 45}{34672458.32} = 13.6 \times 10^{-4} P \text{ MPa}$$



$$\tau_{max} = \left[(2.828 \times 10^{-4} P)^2 + (13.6 \times 10^{-4} P)^2 + 2 \times (2.828 \times 10^{-4} P)(13.6 \times 10^{-4} P) \times 0.724 \right]^{1/2}$$

~~ans~~

$$= 15.77 \times 10^{-4} P$$

$$80 = 15.77 \times 10^{-4} P$$

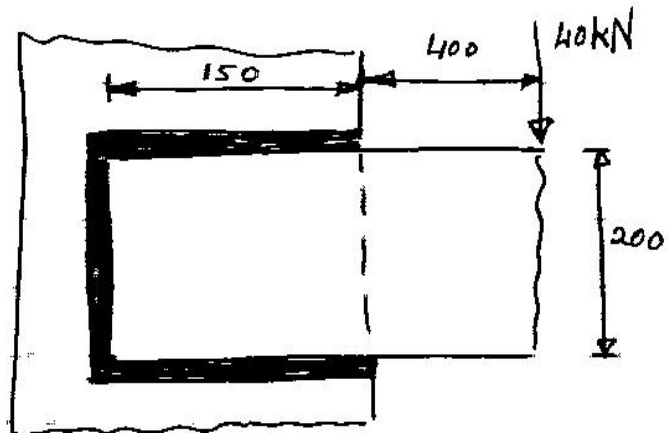
$$P = 50728.55 \text{ N} = 50.728 \text{ kN} \quad \text{ans}$$

12-

$$x = \frac{L^2}{2L+b}$$

$$= \frac{(150)^2}{2 \times 150 + 200}$$

$$= 45 \text{ mm}$$



$$J = 0.707S \left[\frac{(b+2L)^3}{12} - \frac{L^2(b+L)^2}{b+2L} \right]$$

$$= 0.707S \left[\frac{(200+300)^3}{12} - \frac{(150)^2(200+150)}{200+2 \times 150} \right]$$

$$= 3467245.832(S)$$

$$r_2 = [(105)^2 + (100)^2]^{1/2}$$

$$= 145$$

$$\cos \theta = r_1/r_2 = 105/145 = 0.724$$

Direct stress of shear

$$\tau_1 = P/A$$

$$= \frac{40 \times 10^3}{2 \times 0.707S + 0.707S \times 200} = \frac{113.15}{S} \text{ MPa}$$

Secondary stress of shear

$$\tau_2 = \frac{P \cdot e \cdot r_2}{J} = \frac{40 \times 10^3 \times 400 \times 145}{3467245.832S} = \frac{844.763}{S} \text{ MPa}$$

$$\tau_{max} = [\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta]^{1/2}$$

$$= \left[\left(\frac{113.15}{S} \right)^2 + \left(\frac{844.763}{S} \right)^2 + 2 \times \frac{113.15}{S} \times \frac{844.763}{S} \times 0.724 \right]^{1/2}$$

$$= 929/S$$

$$80 = 929/S \Rightarrow S = 11.6 \text{ mm} \Rightarrow t = 8.2 \text{ mm}$$

$$\text{if } e = 400 \text{ then } S = 10 \text{ mm} \Rightarrow t = 7.07 \text{ mm}$$

Solutions of ch. 11 (Screw joints)

1-

From table 11.1 :

stress area for M20 = 245 mm^2 , and for M36 = 817 mm^2

$$P = A \times \sigma_t = 245 \times 200 = 49000 \text{ kN for M20}$$

$$= 817 \times 200 = 163400 = 163.4 \text{ kN for M36}$$

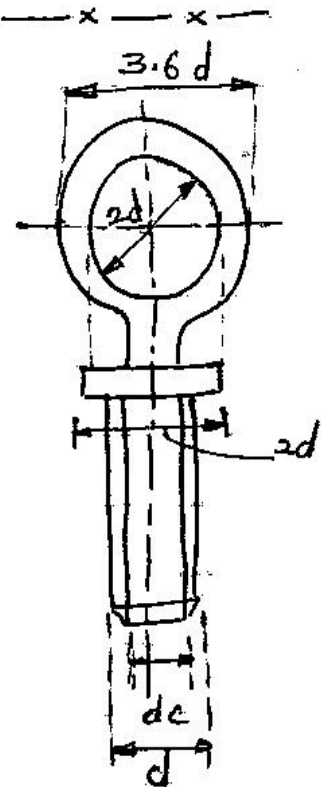
$$2- \quad P = \frac{\pi}{4} (d_c)^2 \cdot \sigma_t$$

$$20 \times 10^3 = \frac{\pi}{4} (d_c)^2 \times 100$$

$$d_c = 15.95 \text{ mm}$$

The nearest value of d_c from table 11.1

$$d_c = 16.933 \text{ mm, M20 ans}$$



$$3- \quad P = \frac{\pi}{4} (D)^2 \cdot p = \frac{\pi}{4} (300)^2 \times 0.7 = 49480.1 \text{ N}$$

$$P = \frac{\pi}{4} d_c^2 \cdot \sigma_t \cdot n$$

$$49480 = \frac{\pi}{4} d_c^2 \times 28 \times 12$$

$$d_c = 13.693$$

The nearest standard value of d_c from table 11.1

$$d_c = 14.933, \quad M = 18$$

4-

$$P = \frac{\pi}{4} (D)^2 \cdot p = \frac{\pi}{4} d_c^2 \cdot \sigma_t \cdot n$$

$$(400)^2 \cdot 1.2 = d_c^2 \cdot 35 \cdot 14$$

$$d_c = 19.794$$

from table 11.1 we obtain standard value of d_c

$$d_c = 20.32 \Rightarrow M24 \quad \text{ans)}$$

— x — x — x — x — x — x — x — x — x —

$$5- \quad P = \frac{\pi}{4} (D)^2 \cdot p = \frac{\pi}{4} (300)^2 \cdot 1 = 70685.835 \text{ N}$$

External Load on each bolt,

$$P_2 = P/n = 70685.835/12 = 5890.486 \text{ N}$$

Tensile Load, $P_i = 2840 \text{ d}$

From table 11.2 we obtain $k = 0.5$ for soft copper gasket

∴ Resultant axial load on the bolt:

$$P_{\text{bolt}} = P_i + k \cdot P_2 = 2840 \text{ d} + 0.5 (5890.486)$$

$$P = \frac{\pi}{4} d_c^2 \cdot \sigma_t = \frac{\pi}{4} (d_c^2) \cdot 100 \quad (d_c = 0.84 \text{ d})$$

$$P_{\text{bolt}} = P$$

$$2840 \text{ d} + 0.5 (5890.486) = \frac{\pi}{4} d_c^2 \cdot 100$$

$$d = 52.26, \quad M56$$

— x — x — x — x — x — x — x — x — x —

$$6- \quad P = \frac{\pi}{4} D^2 \cdot p = \frac{\pi}{4} (300)^2 \cdot 1.5 = 106028.75 \text{ N}$$

$$P_2 = P/n = 106028.75/8 = 13253.6 \text{ N}$$

$$P_i = 2840 \text{ d}, \quad P_{\text{bolt}} = P_i + k \cdot P_2 = 2840 \text{ d} + 0.5 (13253.6)$$

$$P_{\text{bolt}} = \frac{\pi}{4} d_c^2 \cdot \sigma_t$$

$$2840 \text{ d} + 0.5 (13253.6) = \frac{\pi}{4} (0.84 \text{ d})^2 \cdot \sigma_t$$

$d = 20 \text{ mm}$ according to M20 from table 11.1

$$\therefore \sigma_t = 286.13 \text{ MPa}$$

$$7- P = \frac{\pi}{4} D^2 \cdot p = \frac{\pi}{4} (400)^2 \times 1.12 = 140743.35 \text{ N}$$

Assum that : M24 for stud, $d_c = (\text{from table 11.1}) = 20.32 \text{ mm}$

The load effected on studs for (n) number of studs

$$P = \frac{\pi}{4} d_c^2 \times \sigma_t \times n = \frac{\pi}{4} (20.32)^2 \times 35 \times n = 11350.25 n$$

$$n = 140743.35 / 11350.25 = 12.4 \approx 14 \text{ (*)}$$

Taking diamer of stud hole as $d_1 = d + 1 = 24 + 1 = 25 \text{ mm}$

$$D_p = D + 2t + 3d_1 \quad (\text{Assume } t = 10 \text{ mm})$$

$$= 400 + 2 \times 10 + 3 \times 25 = 495 \text{ mm}$$

∴ circumferential Pitch of the stud

$$= \frac{\pi D_p}{n} = \frac{\pi \times 495}{14} = 111.1 \text{ mm}$$

checking the design:

For leak proof joint, the value of circumferential Pitch of studs should be between $20\sqrt{d_1}$ and $30\sqrt{d_1}$

$$\text{Min. circumferential Pitch} = 20\sqrt{d_1} = 20\sqrt{25} = 100 \text{ mm}$$

$$\text{Max. circumferential Pitch} = 30\sqrt{d_1} = 30\sqrt{25} = 150 \text{ mm}$$

Since the value of circumferential Pitch (111.1 mm) Lies between the the min. and max (100 and 150), then the size of studs which we choosen was correct and in safe (satisfactory)

$$\therefore \text{M24, } n = 14$$

ans

(*) - we choose the number of studs (14) as the nearest even even number of (12.4)

نختار اقرب اعل رقم زوجي صحيح

8- Assume that:

$$\sigma_t = 35 \text{ MPa}, t = 10 \text{ mm}, M24 \text{ for studs}$$

from table 11-1, $d_c = 20.32 \text{ mm}$

$$P = \frac{\pi}{4} D^2 \cdot p = \frac{\pi}{4} (400)^2 \cdot 2 = 251327.4 \text{ N}$$

Load effected on studs for (n) number of studs

$$P = \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \cdot n = \frac{\pi}{4} (20.32)^2 \cdot 35 \cdot n = 11350.25 n$$

$$n = 251327.4 / 11350.25 = 22.14$$

The nearest even number of $n = 24$

Taking diameter of stud hole as $d_1 = d + 1 = 24 + 1 = 25 \text{ mm}$

$$D_p = D + 2t + 3d_1 = 400 + 2 \cdot 10 + 3 \cdot 25 = 495 \text{ mm}$$

$$\text{Circumferential Pitch of stud} = \frac{\pi D_p}{n} = \frac{\pi \cdot 495}{24} = 64.8 \text{ mm}$$

checking the design:

For leak proof joint, the value of circumferential Pitch of studs should be lies between $20\sqrt{d_1}$ and $30\sqrt{d_1}$

$$\therefore \text{Min. circumferential Pitch} = 20\sqrt{d_1} = 20\sqrt{25} = 100$$

$$\text{Max. circumferential Pitch} = 30\sqrt{d_1} = 30\sqrt{25} = 150$$

$\therefore$ The design is not in Safe.

Assume that M30, from table 11-1, $d_c = 25.706 \text{ mm}$

$$P = \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \cdot n = \frac{\pi}{4} (25.706)^2 \cdot 35 \cdot n = 251327.4$$

$$n = 13.8 \sim 14 \text{ as the nearest even number}$$

Taking dia. of stud hole as $d_1 = d + 1 = 30 + 1 = 31 \text{ mm}$

$$D_p = D + 2t + 3d_1 = 400 + 2 \cdot 10 + 3 \cdot 31 = 513 \text{ mm}$$

$$C.p = \frac{\pi D_p}{n} = \frac{\pi \cdot 513}{14} = 115.1 \text{ mm}$$

checking:

$$20\sqrt{d_1} = 20\sqrt{31} = 111.35 \text{ mm}$$

$$30\sqrt{d_1} = 30\sqrt{31} = 167 \text{ mm}$$

$\therefore$ the design was in Safe at 115.1

$\therefore$ M30, $n = 14$

ans

10- load acting on cylinder head;

$$P_2 = \frac{\pi}{4} D^2 \cdot p = \frac{\pi}{4} (300)^2 \times 1.5 = 105975 \text{ N}$$

Initial load,

$$P_1 = 1.8 P_2 = 1.8 \times 105975 = 190755 \text{ N}$$

Resultant load (P_{max}) on cylinder head

$$P_{max} = P_1 + k \cdot P_2 = 190755 + 0.5(105975) = 243742.5 \text{ N}$$

$$P_{max} = \text{for one bolt} = \frac{243742.5}{8} = 30467.813 \text{ N}$$

$$P_{min} = \text{for one bolt} = \frac{190755}{8} = 23844.375 \text{ N}$$

$$P_m = \frac{P_{max} + P_{min}}{2} = \frac{30467.813 + 23844.375}{2} = 27156.094 \text{ N}$$

$$P_v = \frac{P_{max} - P_{min}}{2} = \frac{30467.813 - 23844.375}{2} = 3311.719 \text{ N}$$

$$\sigma_m = \frac{P_m}{A} = \frac{P_m}{\frac{\pi}{4} d_c^2} = \frac{4 P_m}{\pi \cdot d_c^2} = \frac{4 \times 27156.094}{\pi \cdot d_c^2} = \frac{34593.75}{d_c^2}$$

$$\sigma_v = \frac{P_v}{A} = \frac{P_v}{\frac{\pi}{4} d_c^2} = \frac{4 P_v}{\pi \cdot d_c^2} = \frac{4 \times 3311.719}{\pi \cdot d_c^2} = \frac{4218.75}{d_c^2}$$

By using Soderberg's Method:

$$\frac{1}{f_s} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v \cdot k_f}{\sigma_e}$$

$$\frac{1}{2} = \frac{34593.75/d_c^2}{350} + \frac{(4218.75 \times 3)/d_c^2}{240}$$

$$d_c = 15.2 \text{ mm}$$

From table 11.1, we obtain that the standard value for

$$d_c = 15.2, \text{ M20 } \underline{\text{ans}}$$

12- load on each stay:

$$P = \text{Area} \times \text{Pressure} = 200 \times 150 \times 1 = 30000 \text{ N/mm}^2$$

The resisting force on the bolts

$$P = \frac{\pi}{4} d_c^2 \cdot \sigma_t$$

$$30000 = \frac{\pi}{4} d_c^2 \times 34$$

$$d_c = 33.52 \text{ mm}$$

from table 11.1, we obtain M39

→ x → x → x → x → x → x → x → x → x →

13- from table 11.1 $d_c = 36.416 \text{ mm}$

$$d = \sqrt{D_o^2 - D_c^2} = \sqrt{(42)^2 - (36.416)^2}$$

$$= 20.925 \text{ mm}$$

→ x → x → x → x → x → x → x → x → x →

14- Direct tensile load on each bolt:

$$W_{t1} = W/n = W/4 = 0.25 W$$

Load in bolt per unit distance:

$$w = \frac{W \cdot L}{2[L_1^2 + L_2^2]} = \frac{W(200)}{2[(80)^2 + (240)^2]} = \frac{W}{640}$$

The heavily loaded bolt at a distance (L_1)

$$W_{t2} = w \cdot L_1 = \frac{W}{640} \times 80 = 0.125 W$$

$$W_{t\text{Total}} = W_{t1} + W_{t2}$$

$$= 0.25 W + 0.125 W = 0.375 W$$

$$W_{t\text{Total}} = \frac{\pi}{4} d_c^2 \cdot \sigma_t$$

$$0.375 W = \frac{\pi}{4} (9.858)^2 \times 60$$

$$W = 12.212 \text{ kN} \quad \underline{\text{ans}}$$

$$15- \quad W_s = \frac{W_{\text{total}}}{n} = \frac{22.5 \times 2 \times 10^3 + 900}{4} = 11475 \text{ N}$$

Let the loads 22.5 kN at the center of the pulley

$$W_A = 22.5 \times 2 \times 10^3 = 45 \times 10^3 \text{ N}$$

$$W_B = 900 \text{ N}$$

The load on belts per unit distance:

$$\omega = \frac{W_A \cdot L_A + W_B \cdot L_B}{2 [L_1^2 + L_2^2]} = \frac{45 \times 10^3 \times 450 + 900 \times 375}{2 [1(75)^2 + (525)^2]} = 36.6 \text{ N/mm}$$

Heavily loaded bolts at the distance L_2 :

$$W_{ts} = \omega \cdot L_2 = 36.6 \times 525 = 19215 \text{ N}$$

$$W_{se} = \frac{1}{2} [(W_t)^2 + 4(W_s)^2]^{1/2}$$

$$= \frac{1}{2} [(19215)^2 + 4(11475)^2]^{1/2} = 14966 \text{ N}$$

$$W_{se} = \frac{\pi}{4} d_c^2 \cdot \tau$$

$$14966 = \frac{\pi}{4} d_c^2 \times 25$$

$$d_c = 27.6 \text{ mm}$$

$$16 - W_s = \frac{W}{n} = \frac{40 \times 10^3}{4} = 10000 \text{ N}$$

the Load on bolts per unit distance

$$w = \frac{W \cdot L}{2 [L_1^2 + L_2^2]} = \frac{40 \times 10^3 \times 600}{2 [(75)^2 + (575)^2]} = 35.69 \text{ N/mm}$$

Heavily loaded bolts at distance L_2

$$W_t = 35.69 \times 575 = 20520.45 \text{ N}$$

$$W_{te} = \frac{W_t}{2} + \frac{1}{2} [(W_t)^2 + 4(W_s)^2]^{1/2}$$

$$= \frac{20520.45}{2} + \frac{1}{2} [(20520.45)^2 + 4(10000)^2]^{1/2}$$

$$= 24587.55 \text{ N}$$

$$W_{te} = \frac{\pi}{4} d_c^2 \cdot \sigma_t$$

$$24587.55 = \frac{\pi}{4} d_c^2 \times 70$$

$$d_c = 21.153$$

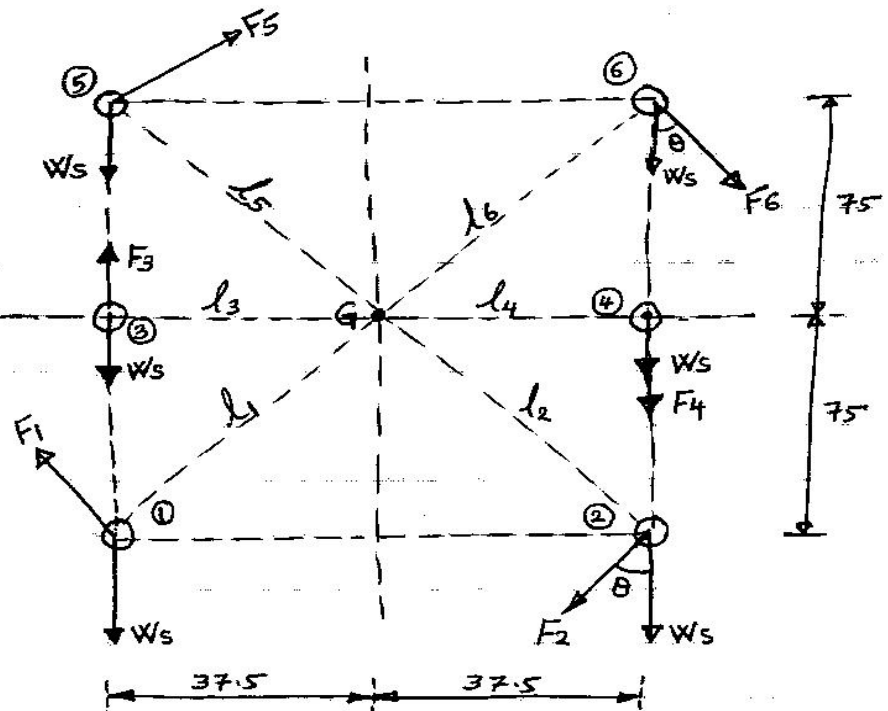
17-

Direct stress

$$W_s = \frac{W}{n} = \frac{50 \times 10^3}{6} = 8333.34 \text{ N}$$

Turning Moment produced by the Load W.

$$T = W \cdot L = 50 \times 10^3 \times 150 = 7500 \times 10^3 \text{ Nmm}$$



$$L_1 = L_2 = L_5 = L_6 = \sqrt{(75)^2 + (37.5)^2} = 83.85 \text{ mm}$$

$$L_3 = L_4 = 37.5 \text{ mm}$$

$$W \cdot L = F_1 \cdot L_1 + F_2 \cdot L_2 + F_3 \cdot L_3 + F_4 \cdot L_4 + F_5 \cdot L_5 + F_6 \cdot L_6$$

$$\text{But } \frac{F_1}{L_1} = \frac{F_2}{L_2} = \dots = \frac{F_n}{L_n}$$

$$\therefore F_2 = F_1 \cdot L_2 / L_1$$

$$\cos \theta_2 = \cos \theta_6 = W_s / F_2 = 8333.34 / 20327.27 = 0.41$$

$$\therefore W \cdot L = \frac{F_1}{L_1} (L_1^2 + L_2^2 + L_3^2 + L_4^2 + L_5^2 + L_6^2)$$

$$7500 \times 10^3 = F_1 / 83.85 (4L_1^2 + 2L_4^2)$$

$$7500 \times 10^3 = F_1 / 83.85 [4 \times (83.85)^2 + 2(37.5)^2]$$

$$F_1 = 20327.27 \text{ N}$$

$$F_1 = F_2 = F_5 = F_6 = 20327.27 \text{ N}$$

$$F_3 = F_4 = F_1 / L_3 / L_1 = 20327.27 \times 37.5 / 83.85 = 9091 \text{ N}$$

From fig. The direct and secondary shear on each bolt, the heavily loaded bolts are (2, 4, 6).

$$\text{The resultant shear: } R_2 = R_6 = [(W_s)^2 + (F_2)^2 + 2W_s \cdot F_2 \cos \theta]^{1/2}$$

$$R_2 = [(8333.34)^2 + (20327.27)^2 + 2 \times 8333.34 \times 20327.27 \times 0.41]^{1/2}$$

$$= 24930.8 \text{ N}$$

$$R_4 = W_s + F_4 = 8333.34 + 9091 = 17424.34 \text{ N}$$

Max Resultant at bolt 6 & 2

$$24930.8 = \frac{\pi}{4} d_c^2 \times 150 \Rightarrow d_c = 14.5 \text{ mm}$$

$$18- \quad W_{t1} = \frac{W}{n} = \frac{2.2 + 4.25}{4} = 1612.5 \text{ N}$$

$$W \cdot L = (\omega \cdot L_1) L_1 \cdot 2 + (\omega \cdot L_2) L_2 \cdot 2$$

$$\omega = \frac{W \cdot L}{2[L_1^2 + L_2^2]} = \frac{6.45 \times 10^3 \times 250}{2[(150)^2 + (200)^2]} = 18.97 \text{ N/mm}$$

$$W_{t2} = \omega \cdot L_2 = 18.97 \times 200 = 3794.12 \text{ N}$$

$$W_t = W_{t1} + W_{t2} = 1612.5 + 3794.12 = \frac{\pi}{4} d_c^2 \times 50$$

$$d_c = 13.546 \text{ mm}$$

— x — x — x — x — x — x — x — x — x —

$$19- \quad L = e - R = 1250 - 300 = 950 \text{ mm}$$

Max. load on bolt:

$$W_t = \frac{2WL(R+r)}{n(2R^2 + r^2)}$$

$$= \frac{2 \times 60 \times 10^3 \times 950(300 + 250)}{4[2 \times (300)^2 + (250)^2]} = 64639.17 \text{ N}$$

$$W_t = \frac{\pi}{4} d_c^2 \cdot \sigma_E$$

$$64639.17 = \frac{\pi}{4} d_c^2 \times 100$$

$$d_c = 28.688 \text{ mm}$$

20 mm

$$\bar{x} = 50 \text{ mm}$$

$$\bar{y} = 60 \text{ mm}$$

for horizontal force

$$W_{t1} = \frac{F_H}{n}$$

$$= \frac{F \sin 60}{5}$$

$$= 7794.23 \text{ N}$$

The horizontal force

causes turning moment

$$\text{counterclockwise : } T_H = F_H \times 35 = 45 \sin 60 \times 35 = 1363950 \text{ N}\cdot\text{mm}$$

$$\text{For vertical force : } W_s = \frac{F \cos 60}{n} = \frac{45 \cos 60}{5} = 4500 \text{ N}$$

Turning Moment due to vertical force

$$T_V = 45 \cos 60 \times (175 + 20) = 4387500 \text{ N}\cdot\text{mm (CCW)}$$

$$\text{Net turning : } 1363950 + 4387500 = 3023550 \text{ N}\cdot\text{mm (CCW)}$$

load on each bolt per unit distance:

$$\omega = \frac{W \cdot L}{2 \left[\frac{L_1^2}{1} + 3 \frac{L_2^2}{2} \right]} = \frac{3023550}{2(50)^2 + 3(150)^2} = 41.7 \text{ N/mm}$$

$$\text{Max load at } L_2 : W_{t2} = \omega \cdot L_2 = 41.7 \times 150 = 6255.62 \text{ N}$$

$$W_{t_{\text{Total}}} = W_{t1} + W_{t2} = 7794.23 + 6255.62 = 14049.85 \text{ N}$$

$$W_{te} = \frac{1}{2} \left[W_t + \left[(W_t)^2 + 4(W_s)^2 \right]^{1/2} \right]$$

$$= \frac{1}{2} \left[14049.85 + \left[(14049.85)^2 + 4(4500)^2 \right]^{1/2} \right] = 15367.56 \text{ N}$$

$$W_{te} = \frac{\pi}{4} d_c^2 \times 70 \Rightarrow d_c = 16.718 \Rightarrow M20$$

$$\text{For } W_{se} = \frac{1}{2} \left[(W_t)^2 + 4(W_s)^2 \right]^{1/2} = \frac{1}{2} \left[(14049.85)^2 + 4(4500)^2 \right]^{1/2} = 8342.6 \text{ N}$$

$$W_{se} = \frac{\pi}{4} d_c^2 \times 70 \Rightarrow d_c = 14.57 \Rightarrow M16$$

